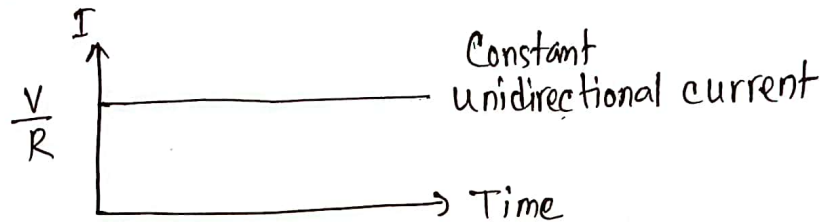


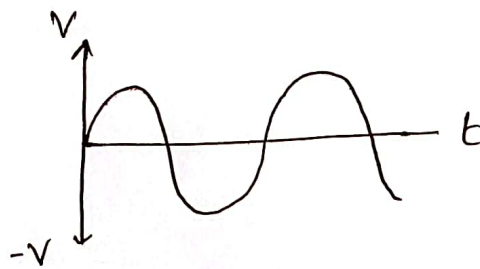
AC Circuit Analysis

Introduction

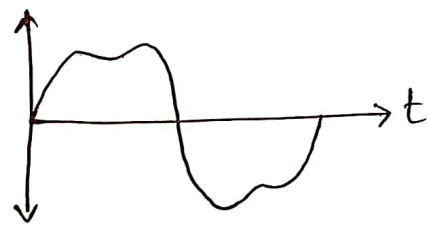
A direct current (dc) is a current which remains constant magnitude with respect to time.



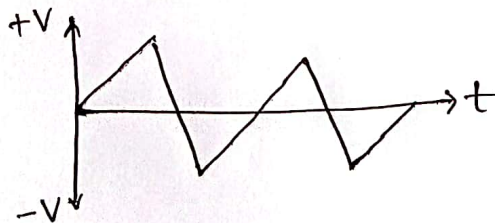
An alternating current (ac) is a current which changes periodically with respect to time both in magnitude and direction.



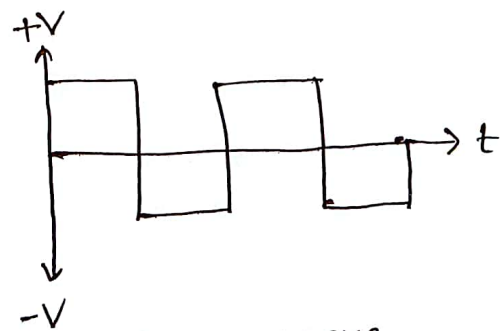
Sine wave



Complex wave



Triangular wave

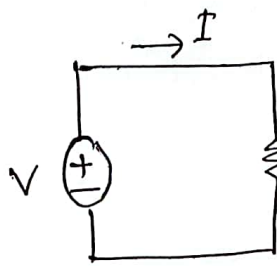


Square wave

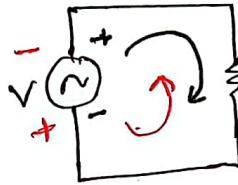
Advantages of AC

- The voltages in ac system can be raised or lowered with the help of transformers. In dc system, raising and lowering of voltages is not so easy.
- Electrical transmission of ac signals at high voltages is possible.
- AC electrical motors are simple in construction, are cheaper and require less maintenance.
- Whenever it is necessary ac supply can be easily converted to obtain dc supply.

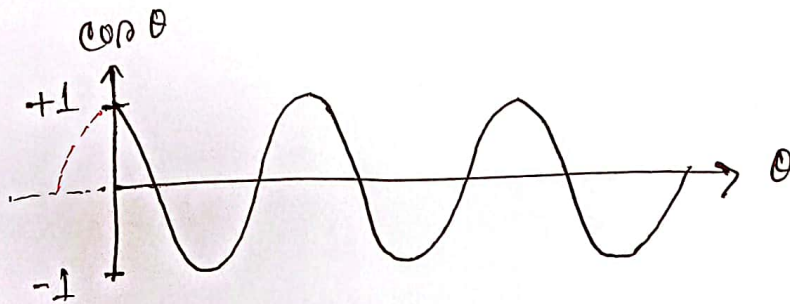
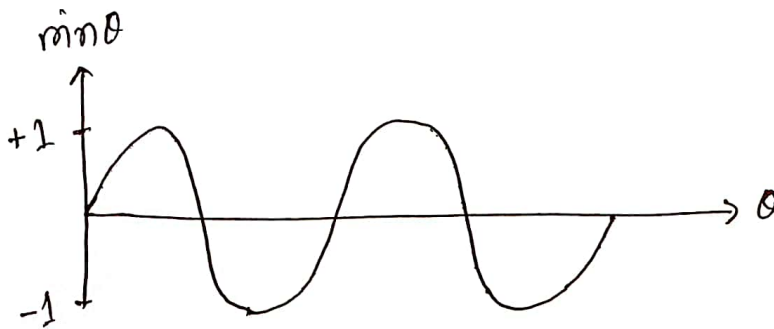
Sinusoidal AC Voltage Sources



DC



AC



$$V(t) = V_m \sin(\omega t + \phi)$$

$$\text{Or, } V(t) = V_m \cos(\omega t + \phi)$$

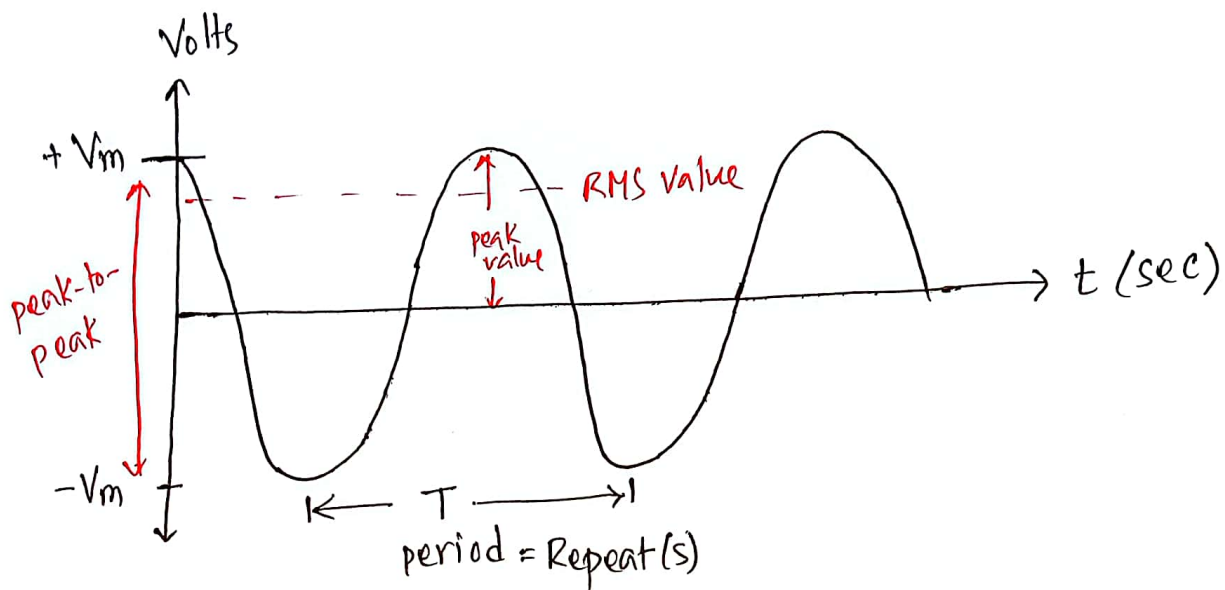
Amplitude

Varies
between
 ± 1

Angular
frequency

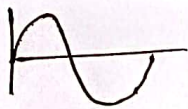
Phase
angle

Sinupoid $\rightarrow$ The shape is like sine or cosine function.

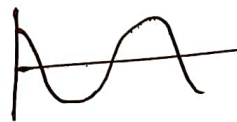


Some Definitions / Terminologies

Waveform: The nature of graph of alternating quantity against time is known as waveform or waveshape.



$$V = V_m \sin(\omega t)$$



$$V = V_m \cos(\omega t)$$

Instantaneous value: The value of an alternating quantity at a particular instant is known as its instantaneous value.

Amplitude: The maximum value attained by an alternating quantity during positive or negative half cycle is called its amplitude.

Cycle: Each repetition of a set of positive and negative instantaneous values of the alternating quantity is called a cycle. Each complete cycle is spread over 360° .

Peak and Peak-to-Peak Values: During each complete cycle of AC signal there are always two maximum or peak values, one for positive half-cycle and the other for the negative half-cycle.

The peak value is measured from zero to the maximum value obtained in either the positive or negative direction. The difference between the peak positive value and the peak negative value is called the peak-to-peak value of the sine wave.

Periodic Time: The time taken by an alternating quantity in seconds to trace one complete cycle is called periodic time or time period. It is usually denoted by symbol T .

Frequency & Angular Frequency:

The number of cycles per second is called frequency and is denoted by symbol f .

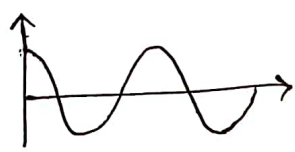
$$\text{Thus, } f = \frac{1}{T} \quad \text{or, } T = \frac{1}{f}.$$

If the angular frequency, ω is expressed in radians per second,

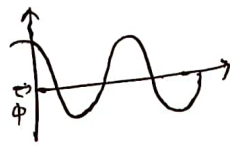
Then, $\omega = \frac{2\pi}{T} = 2\pi f.$

Phase difference

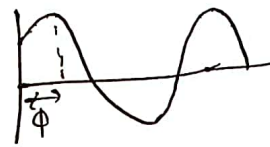
Phase angle, ϕ , tells where we start measuring.



$$\phi = 0$$

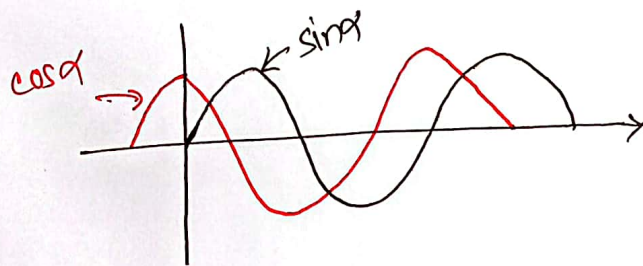


$$\phi > 0$$



$$\phi < 0$$

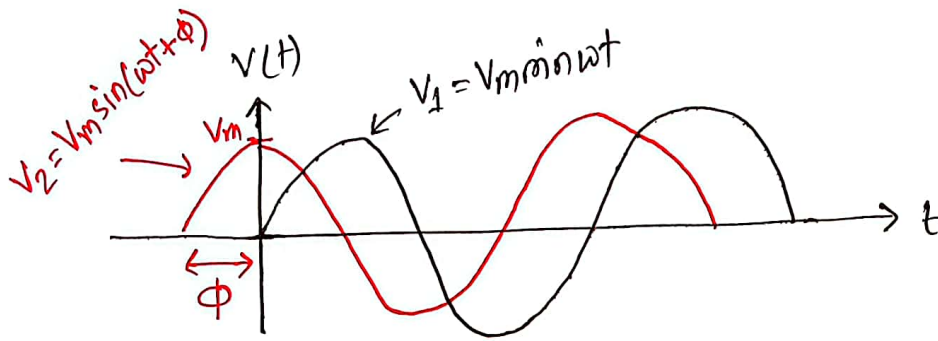
- A periodic function is one that satisfies $v(t) = v(t + nT)$ for all t and for all integers n .
- If phase difference is zero, they are in phase; if phase difference is not zero, they are out of phase.



- The cosine curve is said to lead the sine curve by 90° .
- The sine curve is said to lag the cosine curve by 90° .
- 90° is referred to as the phase angle between two waveforms.

Consider the sinusoidal voltage having phase, ϕ

$$v(t) = V_m \sin(\omega t + \phi)$$



- v_2 leads v_1 by phase ϕ .
- v_1 lags v_2 by phase ϕ .
- v_1 and v_2 are out of phase.

Average Value

The average value of an alternating current is that DC current which transfers across any circuit the same charge as it transferred by that alternating current during the same current time.

For a complete cycle ($0 \rightarrow T$), $I_{avg} = 0$.

For a half-cycle ($0 \rightarrow T/2$), $V_{avg} = 0.637 V_m$

RMS Value (Root mean Square Value)

It is that value of DC current which when flowing through a given circuit for a given time produces the same heat as produced by the alternating current when flowing through the same circuit for the same time.

$$\begin{aligned}\text{RMS value} &= \frac{\text{Max. Value}}{\sqrt{2}} \quad (\text{Only for sine wave}) \\ &= 0.707 \times \text{Max. Value}\end{aligned}$$

$$V_{\text{rms}} = 0.707 V_m ; I_{\text{rms}} = 0.707 I_m.$$

Form Factor

The ratio of the rms value to mean/average value of an alternating quantity is called form factor.

Form factor is used for determining type of wave.

$$\text{Form factor} = \frac{\text{rms value}}{\text{average value}}$$

For alternating current, $i = I_m \sin \omega t$.

$$\text{Form factor} = \frac{I_m / \sqrt{2}}{2I_m / \pi} = 1.11.$$

Peak/Amplitude/Crest Factor

The ratio of maximum value to rms value of an alternating quantity is called amplitude factor.

The physical importance of this factor is in "insulation testing" and "iron lossing" because former depends on maximum value and later depends on maximum flux.

$$\text{Peak factor} = \frac{\text{Maximum value}}{\text{rms value}} = \frac{I_m}{I_m/\sqrt{2}} = 1.41.$$

Problem - 01

Given $i(t) = 5 \cos(400t - 120^\circ)$ A. Determine the period of the current and the frequency in Hertz.

Solution:

$$i(t) = I_m \cos(\omega t + \phi)$$

Hence, $\omega = 400$

$$\Rightarrow 2\pi f = 400 \Rightarrow f = \frac{400}{2\pi} \text{ Hz} = 63.66 \text{ Hz}.$$

$$\begin{aligned} \text{Time period, } T &= \frac{1}{f} = \frac{1}{63.66} = 0.0157 \text{ s} \\ &= 15.71 \text{ ms}. \end{aligned}$$

Problem - 02

Find the amplitude, phase, period and frequency of the sinusoid.

$$v(t) = 12 \cos(50t + 10^\circ).$$

Solution:

The sinusoid

$$v(t) = V_m \cos(\omega t + \phi)$$

The amplitude, $V_m = 12 \text{ V}$

The phase, $\phi = 10^\circ$.

The angular frequency, $\omega = 50 \text{ rad s}^{-1}$.

The period, $T = \frac{2\pi}{\omega} = \frac{2\pi}{50} = 0.1257 \text{ s}$.

The frequency, $f = \frac{1}{T} = \frac{1}{0.1257} = 7.958 \text{ Hz}$.

Problem-03

Given the sinusoid $5 \sin(4\pi t - 60^\circ)$, calculate its amplitude, phase, angular frequency, period, and frequency.

Solution

The sinusoid equation

$$A = A_m \sin(\omega t + \phi)$$

Amplitude, $A_m = 5$.

phase, $\phi = -60^\circ$.

Angular frequency, $\omega = 4\pi \text{ rad s}^{-1}$.

period, $T = \frac{2\pi}{\omega} = \frac{2\pi}{4\pi} = 0.5 \text{ s}$.

frequency, $f = \frac{1}{T} = \frac{1}{0.5} = 2 \text{ Hz}$.

Problem-04

Given the sinusoidal voltage $v(t) = 50 \cos(30t + 10^\circ)$ V.

Find: (a) amplitude, V_m , (b) period, T , (c) frequency, f , and (d) $v(t)$ at $t = 10$ ms.

Solution:

$$v(t) = V_m \cos(\omega t + \phi)$$

a) Amplitude, $V_m = 50$ V.

b) Angular frequency, $\omega = 30 \text{ rad s}^{-1}$.

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{30} = 0.2094 \text{ s} \\ = 209.4 \text{ ms.}$$

c) Frequency, $f = \frac{1}{T} = \frac{1}{0.2094} = 4.775 \text{ Hz}$.

d) At $t = 10 \text{ ms} = 10 \times 10^{-3} \text{ s}$

$$v(t) = 50 \cos(30 \times 10 \times 10^{-3} + 10^\circ) \\ = 49.19 \text{ V.}$$

Problem-05

A current source in a linear circuit has

$$i_s = 8 \cos(500\pi t - 25^\circ) \text{ A.}$$

Find: (a) Amplitude, (b) angular frequency, (c) frequency,
(d) calculate i_s at $t = 2\text{ms}$.

Solution

$$i_s = i_m \cos(\omega t + \phi)$$

a) Amplitude = 8 A

b) Angular frequency, $\omega = 500\pi$

c) frequency, $f = \frac{\omega}{2\pi} = \frac{500\pi}{2\pi} = 250 \text{ Hz.}$

d) At $t = 2\text{ms} = 2 \times 10^{-3} \text{ s.}$

$$i_s = 8 \cos(500\pi \times 2 \times 10^{-3} - 25^\circ)$$
$$= -7.25 \text{ A.}$$

Problem - 06

Given $V = 300 \cos(377t + 30^\circ)$. Find (i) Maximum voltage, (ii) Average voltage, (iii) RMS voltage, (iv) Form factor, (v) Peak factor, (vi) phase, (vii) Frequency, (viii) time period, (ix) wave diagram.

Solution:

i) Maximum voltage $= 300V$

ii) Average voltage $= 0.636 V_m = 0.636 \times 300 = 190.8V$.

iii) RMS value $= \frac{V_m}{\sqrt{2}} = \frac{300}{\sqrt{2}} = 212.13V$.

iv) Form Factor, $K_f = \frac{\text{RMS value}}{\text{Avg. value}} = \frac{212.13}{190.8} = 1.11$.

v) Peak factor, $K_a = \frac{\text{Max. value}}{\text{RMS value}} = \frac{300}{212.13} = 1.414$.

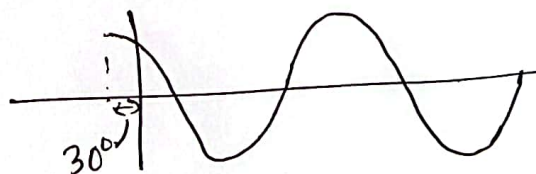
vi) Phase, $\phi = 30^\circ$.

vii) Angular frequency, $\omega = 2\pi f$

$$\Rightarrow f = \frac{\omega}{2\pi} = \frac{377}{2\pi} = 60 \text{ Hz.}$$

viii) Time period, $T = \frac{1}{f} = \frac{1}{60} = 0.0166 \text{ sec.}$

ix) Wave Diagram



Problem-07

A sinusoidal current has a maximum amplitude of 20A. The current passes through one complete cycle in 1ms. The magnitude of the current at zero time is 10A. Find (a) the frequency of the current in Hertz, (b) frequency in radians per second, (c) write the expression for $i(t)$ using cosine function. Express ϕ in degrees, (d) the rms value of the current.

Solution:

a) $T = 1\text{ms}$; Hence, $f = \frac{1}{T} = \frac{1}{1 \times 10^{-3}} = 1000\text{ Hz}$.

b) $\omega = 2\pi f = 2\pi \times 1000 = 2000\pi\text{ rad s}^{-1}$.

c) $i(t) = I_m \cos(\omega t + \phi)$
 $= 20 \cos(2000\pi t + \phi)$

Hence, $i(0) = 10\text{ A}$.

Therefore, $10 = 20 \cos \phi \Rightarrow \phi = \cos^{-1} \frac{10}{20} = 60^\circ$.

Thus, the expression for $i(t)$ becomes

$$i(t) = 20 \cos(2000\pi t + 60^\circ).$$

d) $I_{\text{rms}} = \frac{I_m}{\sqrt{2}} = \frac{20}{\sqrt{2}} = 14.14\text{ A}$.

Problem - 08

A sinusoidal voltage is given by the expression

$$v(t) = 300 \cos(120\pi t + 30^\circ)$$

Find:

- the period of the voltage in milliseconds.
- frequency in Hertz.
- magnitude of v at $t = 2.778 \text{ ms}$.
- rms value of v .

Solution:

- a) From the expression of v ,

$$\omega = 120\pi \Rightarrow \frac{2\pi}{T} = 120\pi$$

$$\Rightarrow T = \frac{2\pi}{120\pi} = \frac{1}{60} = 16.667 \text{ ms.}$$

- b) Frequency, $f = \frac{1}{T} = 60 \text{ Hz}$.

$$\begin{aligned} \text{c) } v &= 300 \cos(120\pi \times 2.778 \times 10^{-3} + 30^\circ) \\ &= 0 \text{ V} \end{aligned}$$

$$\text{d) } v_{\text{rms}} = \frac{300}{\sqrt{2}} = 212.13 \text{ V.}$$

Problem - 09

Consider the sinusoidal voltage

$$v(t) = 80 \cos(1000\pi t - 30^\circ) \text{ V}$$

Find:

- Maximum amplitude of the voltage?
- Frequency in Hertz?
- Frequency in radians per second.
- phase angle in radians.
- phase angle in degrees.
- period in milliseconds.
- calculate the first time after $t=0$ that $v=80\text{V}$.
- the sinusoidal function is shifted $2/3$ ms to the left along the time axis. write the expression for $v(t)$.

Solution:

a) 80V

b) $\omega = 1000\pi \Rightarrow 2\pi f = 1000\pi$

$$\Rightarrow f = \frac{1000\pi}{2\pi} = 500 \text{ Hz.}$$

c) $\omega = 1000\pi = 3141.59 \text{ rad s}^{-1}$.

d) $\varphi (\text{rad}) = -30^\circ = -\frac{\pi}{6} = -0.5236 \text{ rad.}$

e) $\theta = -30^\circ$.

f) $T = \frac{1}{f} = \frac{1}{500} = 2 \text{ ms.}$

g) 80V is the maximum voltage when

$$1000\pi t - \pi/6 = 0 \Rightarrow t = \frac{1}{6000} = 166.67 \mu\text{s.}$$

h) $v = 80 \cos \left[1000\pi \left(t + \frac{0.002}{3} \right) - \frac{\pi}{6} \right]$

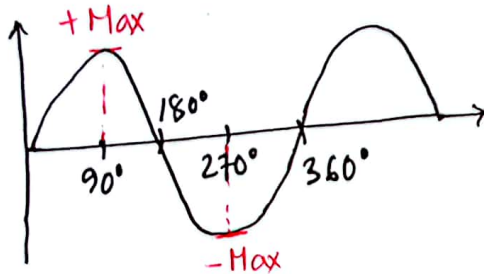
$$= 80 \cos \left[1000\pi t + \frac{2\pi}{3} - \frac{\pi}{6} \right]$$

$$= 80 \cos [1000\pi t + \pi/2]$$

$$= -80 \sin 1000\pi t \text{ V.}$$

Trigonometric Identities

Sinusoidal



(+) sine wave

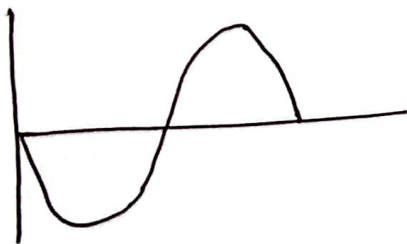
$$\sin 0^\circ = 0$$

$$\sin 90^\circ = 1$$

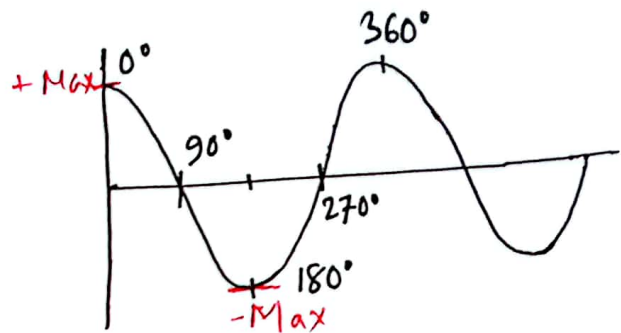
$$\sin 180^\circ = 0$$

$$\sin 270^\circ = -1$$

$$\sin 360^\circ = 0$$



(-) sine wave



(+) cosine wave

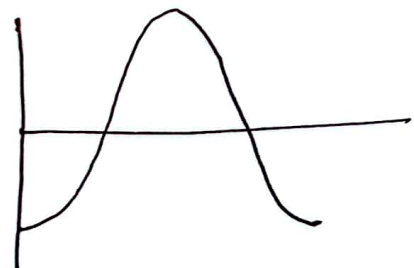
$$\cos 0^\circ = 1$$

$$\cos 90^\circ = 0$$

$$\cos 180^\circ = -1$$

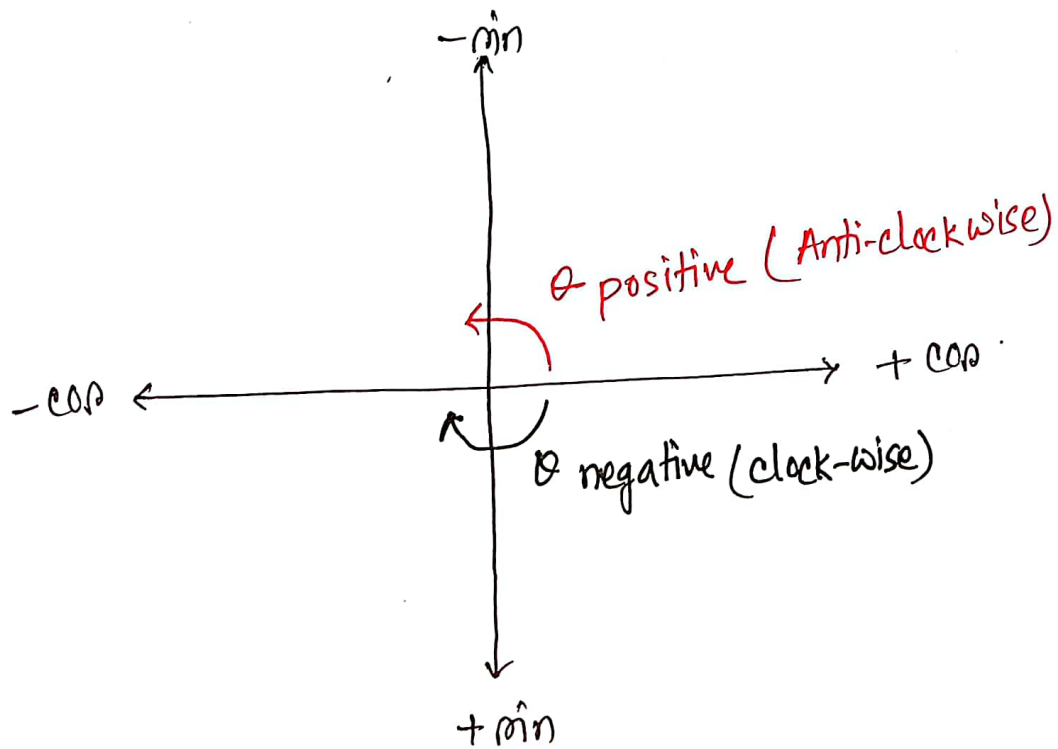
$$\cos 270^\circ = 0$$

$$\cos 360^\circ = 1$$



(-) cosine wave

Graphically relating sine and cosine functions



$$\sin(\omega t \pm 180^\circ) = -\sin \omega t$$

$$\cos(\omega t \pm 180^\circ) = -\cos \omega t$$

$$\sin(\omega t + 90^\circ) = \cos \omega t$$

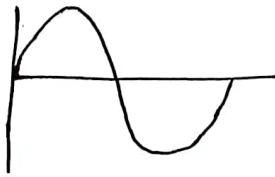
$$\sin(\omega t - 90^\circ) = -\cos \omega t$$

$$\cos(\omega t + 90^\circ) = -\sin \omega t$$

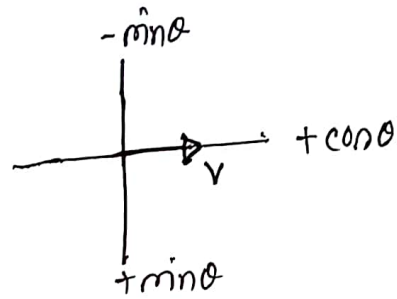
$$\cos(\omega t - 90^\circ) = \sin \omega t.$$

Draw the wave diagram and vector diagram

1) $V = 200 \cos \omega t$

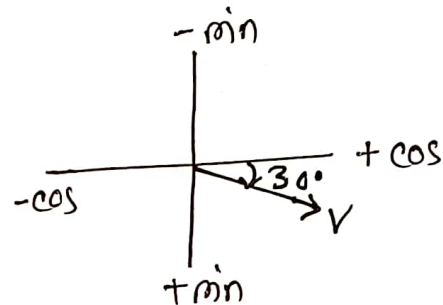
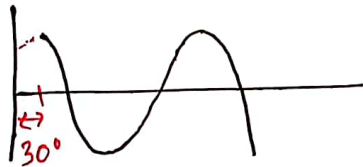


wave diagram

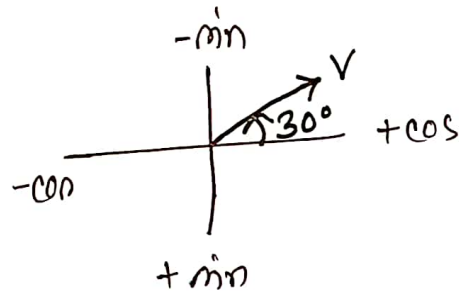
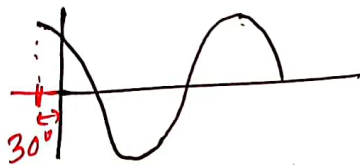


vector diagram

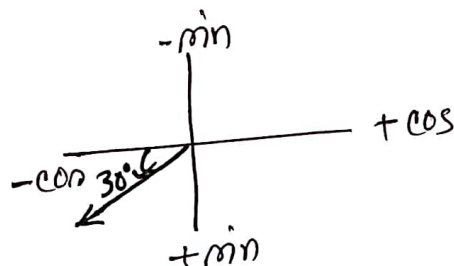
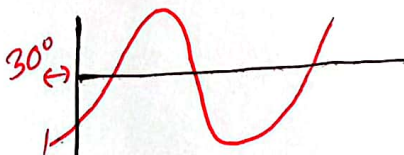
2) $V = 220 \cos (\omega t - 30^\circ)$



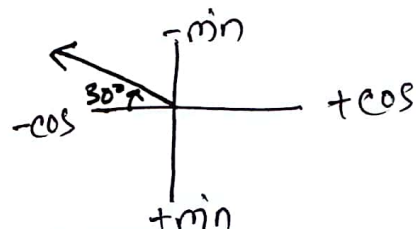
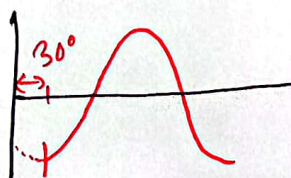
3) $V = 210 \cos (\omega t + 30^\circ)$



4) $-100 \cos (\omega t + 30^\circ)$

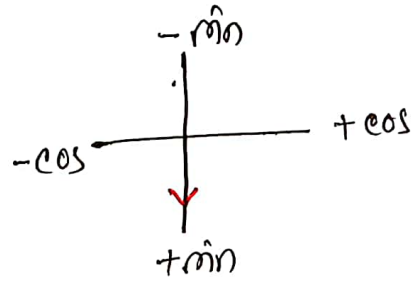
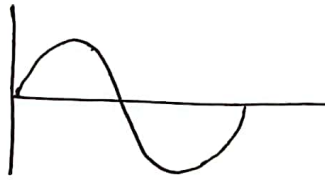


5) $-100 \cos (\omega t - 30^\circ)$

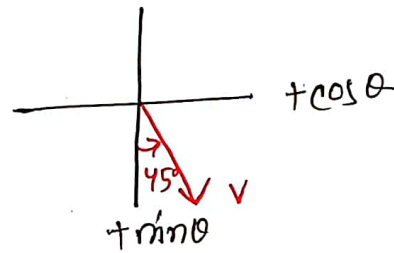
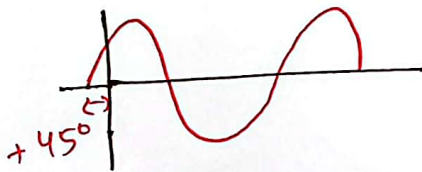


Draw the wave diagram and vector diagram

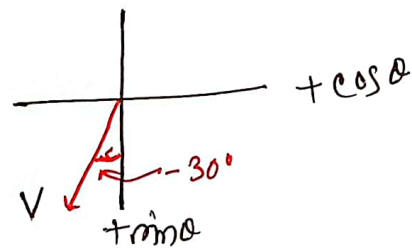
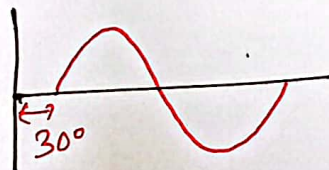
1) $200 \sin \omega t$



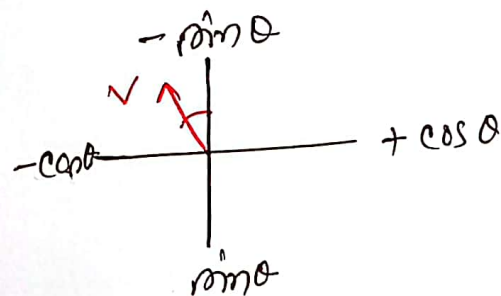
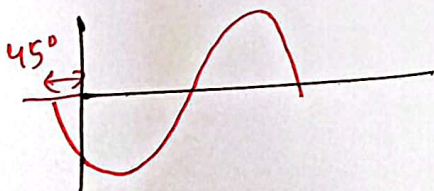
2) $V = 200 \sin(\omega t + 45^\circ)$



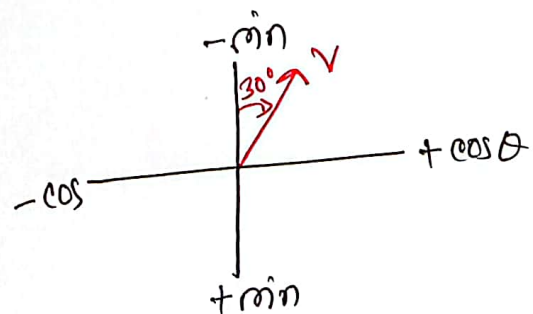
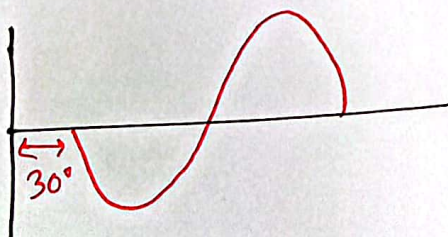
3) $V = 200 \sin(\omega t - 30^\circ)$



4) $V = -200 \sin(\omega t + 45^\circ)$

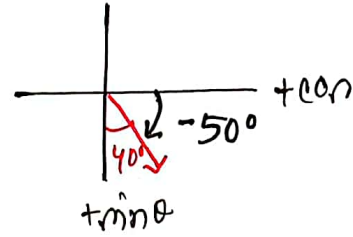


5) ~~V =~~ $V = -200 \sin(\omega t - 30^\circ)$

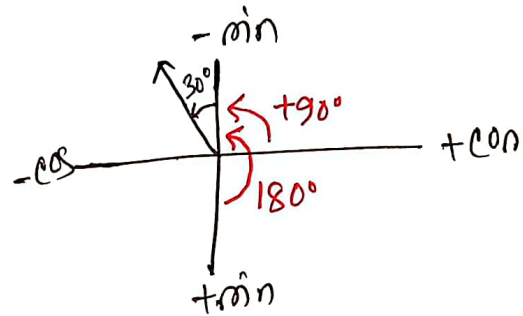


Conversion (sine to cos / cos to sine)

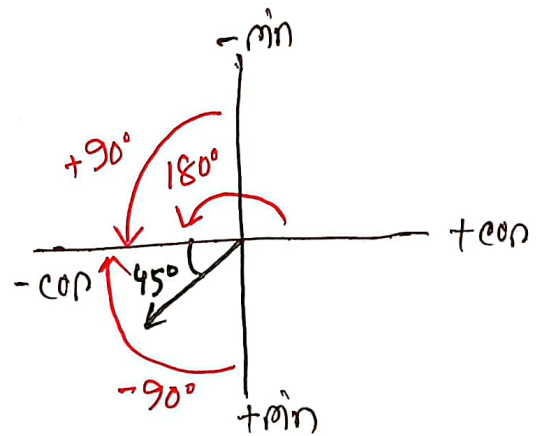
$$\begin{aligned} 1) & 200 \sin(\omega t + 40^\circ) \\ &= 200 \cos(\omega t + 40^\circ - 90^\circ) \\ &= 200 \cos(\omega t - 50^\circ) \end{aligned}$$



$$\begin{aligned} 2) & -100 \sin(\omega t + 30^\circ) \\ &= 100 \cos(\omega t + 30^\circ + 90^\circ) \\ &= 100 \cos(\omega t + 120^\circ) \\ &\Rightarrow 100 \sin(\omega t + 30^\circ + 180^\circ) \\ &= 100 \sin(\omega t + 210^\circ) \end{aligned}$$



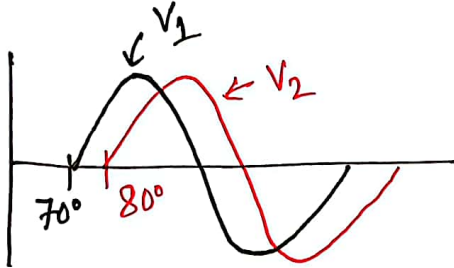
$$\begin{aligned} 3) & -100 \cos(\omega t + 45^\circ) \\ &= 100 \cos(\omega t + 45^\circ + 180^\circ) \\ &= 100 \cos(\omega t + 225^\circ) \\ &\Rightarrow 100 \sin(\omega t + 45^\circ - 90^\circ) \\ &= 100 \sin(\omega t - 45^\circ) \\ &\Rightarrow -100 \sin(\omega t + 45^\circ + 90^\circ) \\ &= -100 \sin(\omega t + 135^\circ) \end{aligned}$$



Leading and Lagging

$$V_1 = V_m \sin(\omega t - 70^\circ)$$

$$V_2 = V_m \sin(\omega t - 80^\circ)$$



V_1 leads V_2 by 10° .

V_2 lags V_1 by 10° .

Problem - 10

Determine the relative phase relationship of the two waves.

$$i(t) = 5 \sin(377t - 20^\circ) \text{ V}$$

$$v(t) = 10 \cos(377t + 30^\circ) \text{ V}$$

Solution:

$$i(t) = 5 \sin(377t - 20^\circ) \text{ V}$$

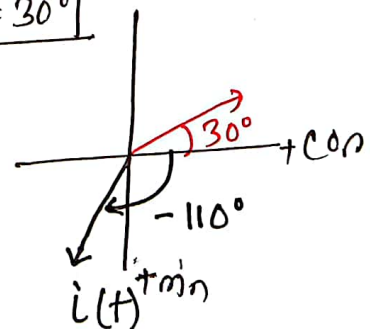
$$= 5 \cos(377t - 20^\circ - 90^\circ) \text{ V}$$

$$= 5 \cos(377t - 110^\circ) \text{ V}; \quad \boxed{\theta_i = -110^\circ}$$

$$v(t) = 10 \cos(377t + 30^\circ) \text{ V}; \quad \boxed{\theta_v = 30^\circ}$$

$$\begin{aligned} \text{The phase difference} &= \theta_v - \theta_i \\ &= 140^\circ. \end{aligned}$$

θ_v leads θ_i by 140° .



Problem - 11(a)

Express the following functions in cosine form.

a) $4 \sin(\omega t - 30^\circ)$

b) $-2 \sin 6t$

c) $-10 \sin(\omega t + 20^\circ)$

Solution:

a) $4 \sin(\omega t - 30^\circ)$

$$= 4 \cos(\omega t - 30^\circ - 90^\circ)$$

$$= 4 \cos(\omega t - 120^\circ).$$

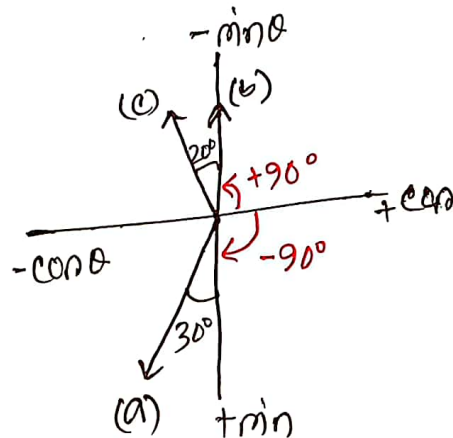
b) $-2 \sin 6t$

$$= 2 \cos(6t + 90^\circ)$$

c) $-10 \sin(\omega t + 20^\circ)$

$$= 10 \cos(\omega t + 20^\circ + 90^\circ)$$

$$= 10 \cos(\omega t + 110^\circ)$$



Problem-11 (b)

Calculate the phase angle between $v_1 = -10 \cos(\omega t + 50^\circ)$ and $v_2 = 12 \sin(\omega t - 10^\circ)$. State which sinusoid is leading.

Solution:

$$\begin{aligned} v_1 &= -10 \cos(\omega t + 50^\circ) \\ &= 10 \cos(\omega t + 50^\circ + 180^\circ) \\ &= 10 \cos(\omega t + 230^\circ) \end{aligned}$$

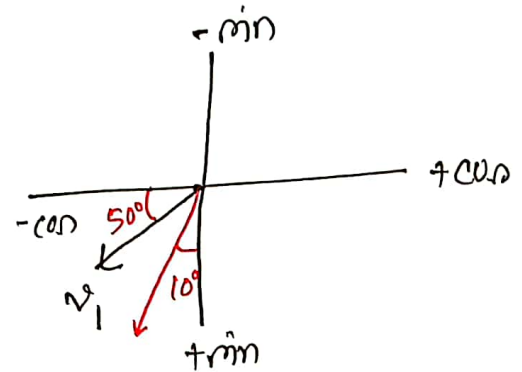
$$\boxed{\therefore \theta_{v_1} = 230^\circ}$$

$$\begin{aligned} v_2 &= 12 \sin(\omega t - 10^\circ) \\ &= 12 \cos(\omega t - 10^\circ + 270^\circ) \\ &= 12 \cos(\omega t + 260^\circ) \end{aligned}$$

$$\boxed{\therefore \theta_{v_2} = 260^\circ}$$

The phase angle between v_1 and v_2 is $\theta_{v_2} - \theta_{v_1} = 30^\circ$.

Hence, v_2 leads v_1 by 30° .



Problem-12

Find the phase angle between

$$i_1 = -4 \sin(377t + 25^\circ)$$

$$\text{and } i_2 = 5 \cos(377t - 40^\circ)$$

Does i_1 lead or lag i_2 ?

Solution:

$$\begin{aligned} i_1 &= -4 \sin(377t + 25^\circ) \\ &= 4 \cos(377t + 25^\circ + 90^\circ) \\ &= 4 \cos(377t + 115^\circ) \end{aligned}$$

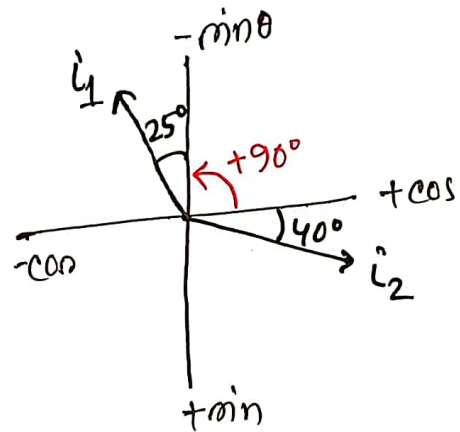
$$\boxed{\theta_{i_1} = 115^\circ}$$

$$i_2 = 5 \cos(377t - 40^\circ)$$

$$\boxed{\theta_{i_2} = -40^\circ}$$

$$\begin{aligned} \text{The phase angle between } i_1 \text{ and } i_2 &= \theta_{i_1} - \theta_{i_2} \\ &= 115^\circ - (-40^\circ) \\ &= 155^\circ. \end{aligned}$$

i_1 leads i_2 by 155° .



Phasors

Phasor

A Phasor is a complex number and represents as a vector in complex plane, that has amplitude and phase of a sinusoid.

$$V(t) = V_m \cos(\omega t + \theta) \longrightarrow V_m \angle \theta.$$

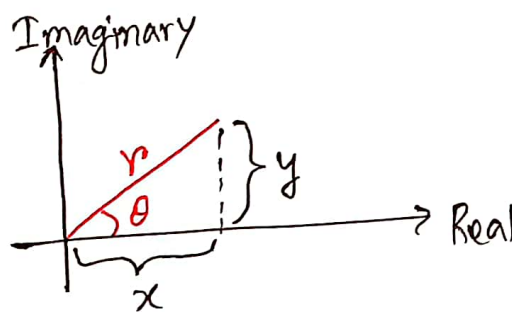
Amplitude Angular frequency phase

Complex Number

$$z = x + jy$$

real imaginary

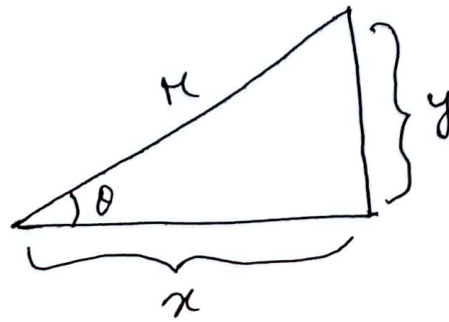
Complex number = real number + imaginary number



Rectangular form: $z = x + jy$

Polar form : $z = r \angle \theta$

Exponential form: $z = r e^{j\theta}$



$$r = \sqrt{x^2 + y^2}$$

$$\tan \theta = \frac{y}{x}$$

$$\theta = \tan^{-1} \frac{y}{x}$$

$$\cos \theta = \frac{x}{r}$$

$$\Rightarrow x = r \cos \theta$$

$$\sin \theta = \frac{y}{r}$$

$$\Rightarrow y = r \sin \theta$$

Rectangular to Polar

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \tan^{-1} \frac{y}{x}$$

Polar to Rectangular

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = x + jy$$

$$= r \cos \theta + j r \sin \theta$$

$$= r (\cos \theta + j \sin \theta)$$

$$\boxed{z = r e^{j\theta}}$$

$$\left[\text{Euler's Identity} \right]$$

$$e^{\pm j\theta} = \cos \theta \pm j \sin \theta$$

Operations:

Addition:

$$Z = a + jb ; W = c + jd$$

$$Z + W = (a + jb) + (c + jd) = (a + c) + j(b + d)$$

Subtraction:

$$Z - W = (a + jb) - (c + jd) = (a - c) + j(b - d)$$

Multiplication:

$$r_z = \sqrt{a^2 + b^2} ; \theta_z = \tan^{-1} b/a$$

$$r_w = \sqrt{c^2 + d^2} ; \theta_w = \tan^{-1} d/c$$

$$Z \cdot W = (r_z \angle \theta_z) (r_w \angle \theta_w) = r_z r_w \angle (\theta_z + \theta_w)$$

Division:

$$\frac{Z}{W} = \frac{r_z \angle \theta_z}{r_w \angle \theta_w} = \frac{r_z}{r_w} \angle (\theta_z - \theta_w)$$

Exponentiation:

$$Z^n = (r_z \angle \theta_z)^n = r_z^n \angle (n\theta_z)$$

Complex Conjugate:

$$z^* = a - jb = r_z \angle -\theta_z.$$

$$w^* = c - jd = r_w \angle -\theta_w.$$

Euler's Identity

$$e^{\pm j\theta} = \cos(\theta) \pm j \sin(\theta)$$

$$\cos \theta = \operatorname{Re}(e^{j\theta}); \sin \theta = \operatorname{Im}(e^{j\theta}).$$

Sinusoid to Phasor Form

$$v(t) = V_m \cos(\omega t + \phi) \Leftrightarrow V = V_m \angle \phi.$$

(Time-domain representation)

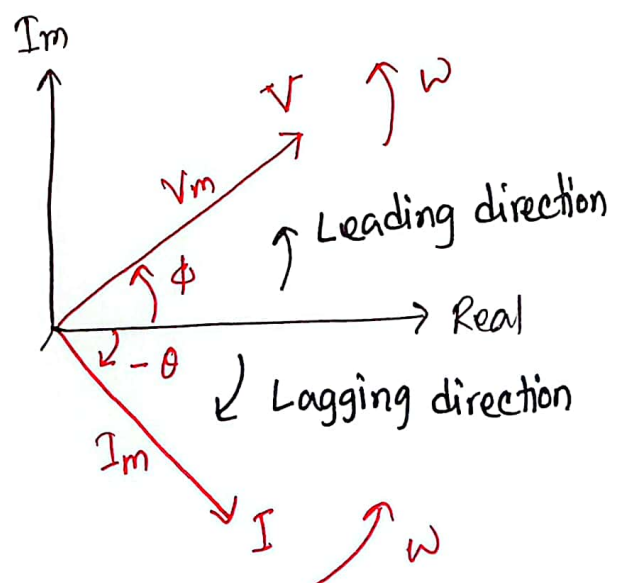
(Phasor-domain representation)

Graphical Representation of a Phasor

$$v(t) = V_m \cos(\omega t + \phi) \Leftrightarrow V = V_m \angle \phi.$$



$$I = I_m \angle -\theta$$



Sinusoid-phasor transformation

Time domain

Phasor domain

$$V_m \cos(\omega t + \phi) \longrightarrow V_m \angle \phi$$

$$\begin{aligned} V_m \sin(\omega t + \phi) \\ = V_m \cos(\omega t + \phi - 90^\circ) \end{aligned} \longrightarrow V_m \angle \phi - 90^\circ.$$

$$I_m \cos(\omega t + \phi) \longrightarrow I_m \angle \phi$$

$$\begin{aligned} I_m \sin(\omega t + \phi) \\ = I_m \cos(\omega t + \phi - 90^\circ) \end{aligned} \longrightarrow I_m \angle \phi - 90^\circ.$$

Problem-13

Evaluate these complex numbers:

$$a) (40 \angle 50^\circ + 20 \angle -30^\circ)^{1/2}.$$

$$b) \frac{10 \angle -30^\circ + (3 - j4)}{(2 + j4)(3 - j5)^*}.$$

$$c) [(5 + j2)(-1 + j4) - 5 \angle 60^\circ]^*.$$

$$d) \frac{10 + j5 + 3 \angle 40^\circ}{-3 + j4} + 10 \angle 30^\circ + j5.$$

Solution

$$a) (40 \angle 50^\circ + 20 \angle -30^\circ)^{1/2}$$

$$= ((25.71 + j30.64) + (17.32 - j10))^{1/2}$$

$$= (43.03 + j20.64)^{1/2}$$

$$= (47.72 \angle 25.63^\circ)^{1/2}$$

$$= 6.91 \angle 12.81^\circ$$

$$b) \frac{10 \angle -30^\circ + (3 - j4)}{(2 + j4)(3 - j5)^*} = \frac{(8.66 - j5) + (3 - j4)}{(2 + j4)(3 + j5)}$$
$$= \frac{11.66 - j9}{(4.47 \angle 63.43^\circ)(5.83 \angle 59.04^\circ)}$$
$$= \frac{14.73 \angle -37.66^\circ}{26.06 \angle 122.47^\circ}$$

$$= 0.565 \angle -160.13^\circ$$

$$c) [(5 + j2)(-1 + j4) - 5 \angle 60^\circ]^*$$
$$= [(5.385 \angle 21.8^\circ)(4.123 \angle 104.03^\circ) - 5 \angle 60^\circ]^*$$
$$= [22.20 \angle 125.83^\circ - 5 \angle 60^\circ]^*$$
$$= [-13 + j18 - 2.5 - j4.33]^*$$
$$= [-15.5 + j13.67]^* = -15.5 - j13.67$$

$$\begin{aligned}
 (d) \quad & \frac{10 + j5 + 3 \angle 40^\circ}{-3 + j4} + 10 \angle 30^\circ + j5 \\
 = & \frac{10 + j5 + 2.298 + j1.928}{-3 + j4} + 10 \angle 30^\circ + j5 \\
 = & \frac{12.298 + j6.928}{-3 + j4} + 10 \angle 30^\circ + j5 \\
 = & \frac{14.115 \angle 29.39^\circ}{5 \angle 126.87^\circ} + 10 \angle 30^\circ + j5 \\
 = & 2.823 \angle -97.48^\circ + 10 \angle 30^\circ + j5 \\
 = & -0.367 - j2.8 + 8.66 + j5 + j5 \\
 = & 8.293 + j7.2
 \end{aligned}$$

Problem - 14

Calculate these complex numbers and express your results in rectangular form.

$$a) \frac{15 \angle 45^\circ}{3 - j4} + j2$$

$$b) \frac{8 \angle -20^\circ}{(2+j)(3-j4)} + \frac{10}{-5+j12}$$

$$c) 10 + (8 \angle 50^\circ)(5 - j12).$$

Solution:

$$\begin{aligned} a) \frac{15 \angle 45^\circ}{3 - j4} + j2 &= \frac{15 \angle 45^\circ}{5 \angle -53.13^\circ} + j2 \\ &= 3 \angle 98.13^\circ + j2 \\ &= -0.4245 + j2.97 + j2 \\ &= -0.4243 + j4.97, \end{aligned}$$

$$\begin{aligned} b) \frac{8 \angle -20^\circ}{(2+j)(3-j4)} + \frac{10}{-5+j12} &= \frac{8 \angle -20^\circ}{(2.23 \angle 26.565^\circ)(5 \angle -53.13^\circ)} + \frac{10}{13 \angle 112.62^\circ} \\ &= \frac{8 \angle -20^\circ}{11.15 \angle -26.565^\circ} + \frac{10}{13 \angle 112.62^\circ} \\ &= 0.717 \angle 6.565^\circ + 0.769 \angle -112.62^\circ \\ &= 0.712 + j0.08 + (-0.296 - j0.71) \\ &= 0.416 - j0.63, \end{aligned}$$

$$\begin{aligned}
 (c) \quad & 10 + (8 \angle 50^\circ)(5 - j12) \\
 & = 10 + (8 \angle 50^\circ)(13 \angle -67.38^\circ) \\
 & = 10 + 104 \angle -17.38^\circ \\
 & = 10 + 99.25 - j31.07 \\
 & = 109.25 - j31.07.
 \end{aligned}$$

Problem-15

Evaluate the following complex numbers and leave your results in polar form.

$$a) \quad 5 \angle 30^\circ \left(6 - j8 + \frac{3 \angle 60^\circ}{2 + j} \right)$$

$$b) \quad \frac{(10 \angle 60^\circ)(35 \angle -50^\circ)}{(2 + j6) - (5 + j)}$$

Solution:

$$a) \quad 5 \angle 30^\circ \left(6 - j8 + \frac{3 \angle 60^\circ}{2 + j} \right)$$

$$= 5 \angle 30^\circ (6 - j8 + 1.119 + j0.739)$$

$$= 5 \angle 30^\circ (10.168 \angle -45.57^\circ)$$

$$= 50.84 \angle -15.565^\circ.$$

$$\begin{aligned}
 (b) \quad \frac{(10 \angle 60^\circ)(35 \angle -50^\circ)}{(2+j6) - (5+j)} &= \frac{350 \angle 10^\circ}{5.83 \angle 120.96^\circ} \\
 &= 60.034 \angle -110.96^\circ.
 \end{aligned}$$

Problem - 16

Transform these sinusoids to phasors:

(a) $i = 6 \cos(50t - 40^\circ) \text{ A}$

(b) $v = -4 \sin(30t + 50^\circ) \text{ V}$

(c) $-10 \cos(4t + 75^\circ)$

(d) $5 \sin(20t - 10^\circ)$

(e) $4 \cos 2t + 3 \sin 2t$

(f) $v(t) = 21 \cos(4t - 15^\circ) \text{ V}$

(g) $i(t) = -8 \sin(10t + 70^\circ) \text{ mA}$

(h) $v(t) = 120 \sin(10t - 50^\circ) \text{ V}$

(i) $i(t) = -60 \cos(30t + 10^\circ) \text{ mA}.$

Solution:

$$(a) \ i = 6 \cos(50t - 40^\circ) \text{ A}$$

$$I = 6 \angle -40^\circ.$$

$$(b) \ v = -4 \sin(30t + 50^\circ)$$

$$= 4 \cos(30t + 50^\circ + 90^\circ)$$

$$= 4 \cos(30t + 140^\circ)$$

$$V = 4 \angle 140^\circ.$$

$$(c) \ -10 \cos(4t + 75^\circ) = 10 \cos(4t + 75^\circ + 180^\circ)$$

$$= 10 \cos(4t + 255^\circ)$$

$$= 10 \angle 255^\circ.$$

$$\text{or, } -10 \cos(4t + 75^\circ) = 10 \cos(4t + 75^\circ - 180^\circ)$$

$$= 10 \cos(4t - 105^\circ)$$

$$= 10 \angle -105^\circ.$$

$$(d) \ 5 \sin(20t - 10^\circ) = 5 \cos(20t - 10^\circ - 90^\circ)$$

$$= 5 \cos(20t - 100^\circ)$$

$$= 5 \angle -100^\circ.$$

$$(e) \ 4 \cos 2t + 3 \sin 2t = 4 \cos 2t + 3 \cos(2t - 90^\circ)$$

$$= 4 \angle 0^\circ + 3 \angle -90^\circ.$$

$$= 4 - j3$$

$$= 5 \angle -36.87^\circ.$$

$$(f) v(t) = 21 \cos(4t - 15^\circ) \text{ V}$$

$$\underline{V} = 21 \angle -15^\circ.$$

$$(g) i(t) = -8 \sin(10t + 70^\circ) \text{ mA}$$

$$= 8 \cos(10t + 70^\circ + 90^\circ) \text{ mA}$$

$$= 8 \cos(10t + 160^\circ) \text{ mA}$$

$$\underline{I} = 8 \angle 160^\circ \text{ mA}.$$

$$(h) v(t) = 120 \sin(10t - 50^\circ) \text{ V}$$

$$= 120 \cos(10t - 50^\circ - 90^\circ) \text{ V}$$

$$= 120 \cos(10t - 140^\circ) \text{ V}$$

$$\underline{V} = 120 \angle -140^\circ \text{ V}.$$

$$(i) i(t) = -60 \cos(30t + 10^\circ) \text{ mA}$$

$$= 60 \cos(30t + 10^\circ + 180^\circ) \text{ mA}$$

$$= 60 \cos(30t + 190^\circ) \text{ mA}$$

$$\underline{I} = 60 \angle 190^\circ \text{ mA}.$$

Problem-17

Find the sinusoids represented by these phasors.

(a) $I = -3 + j4 \text{ A}$.

(b) $V = -10 \angle 30^\circ \text{ V}$

(c) $I = j(5 - j12) \text{ A}$

(d) $V_1 = 60 \angle 15^\circ \text{ V}$, $\omega = 1$.

(e) $V_2 = 6 + j8 \text{ V}$, $\omega = 40$.

(f) $I_1 = -0.5 - j1.2 \text{ A}$, $\omega = 10^3$.

Solution:

(a) $I = -3 + j4$

$$= 5 \angle 126.87^\circ$$

$$i(t) = 5 \cos(\omega t + 126.87^\circ)$$

$$\begin{aligned} \text{(b) } V &= -10 \angle 30^\circ \text{ V} = -10 \cos(\omega t + 30^\circ) \text{ V} \\ &= 10 \cos(\omega t + 30^\circ - 180^\circ) \text{ V} \end{aligned}$$

$$v(t) = 10 \cos(\omega t - 150^\circ) \text{ V}.$$

$$\text{Or, } V = -10 \angle 30^\circ = 10 \cos(\omega t + 30^\circ + 180^\circ)$$

$$v(t) = 10 \cos(\omega t + 210^\circ) \text{ V}.$$

$$(c) \mathbf{I} = \hat{j}(5 - j12) \text{ A} = \hat{j}5 + 12 = 13 \angle 22.62^\circ.$$

$$i(t) = 13 \cos(\omega t + 22.62^\circ) \text{ A}.$$

$$(d) \mathbf{V}_1 = 60 \angle 15^\circ \text{ V}, \omega = 1.$$

$$v_1(t) = 60 \cos(\omega t + 15^\circ)$$

$$= 60 \cos(t + 15^\circ) \quad [\because \omega = 1]$$

$$(e) \mathbf{V}_2 = 6 + j8 \text{ V}, \omega = 40$$

$$= 10 \angle 53.13^\circ \text{ V}$$

$$v_2(t) = 10 \cos(40t + 53.13^\circ) \text{ V}$$

$$(f) \mathbf{I}_1 = -0.5 - j1.2 \text{ A}, \omega = 10^3.$$

$$= 1.3 \angle -112.62^\circ.$$

$$= 1.3 \cos(10^3 t - 112.62^\circ).$$

Problem - 18

Using phasors, find

$$(a) 3 \cos(20t + 10^\circ) - 5 \cos(20t - 30^\circ)$$

$$(b) 40 \sin 50t + 30 \cos(50t - 45^\circ)$$

$$(c) 20 \sin 400t + 10 \cos(400t + 60^\circ) - 5 \sin(400t - 20^\circ).$$

Solution:

$$(a) 3 \cos(20t + 10^\circ) - 5 \cos(20t - 30^\circ)$$

$$= 3 \angle 10^\circ - 5 \angle -30^\circ.$$

$$= 2.954 + j 0.52 - 4.33 + j 2.5$$

$$= -1.376 + j 3.02$$

$$= 3.32 \angle 114.5^\circ.$$

$$= 3.32 \cos(20t + 114.5^\circ).$$

$$(b) 40 \sin 50t + 30 \cos(50t - 45^\circ)$$

$$= 40 \cos(50t - 90^\circ) + 30 \cos(50t - 45^\circ)$$

$$= 40 \angle -90^\circ + 30 \angle -45^\circ.$$

$$= -j 40 + 21.21 - j 21.21$$

$$= 21.21 - j 61.21$$

$$= 64.78 \angle -70.89^\circ.$$

$$= 64.78 \cos(50t - 70.89^\circ).$$

$$(c) 20 \sin 400t + 10 \cos(400t + 60^\circ) - 5 \sin(400t - 20^\circ)$$

$$= 20 \cos(400t - 90^\circ) + 10 \cos(400t + 60^\circ) + 5 \cos(400t - 20^\circ + 90^\circ)$$

$$= 20 \angle -90^\circ + 10 \angle 60^\circ + 5 \angle 70^\circ.$$

$$= -j 20 + 5 + j 8.66 + 1.71 + j 4.69$$

$$= 6.71 - j6.65$$

$$= 9.44 \angle -44.74^\circ.$$

$$= 9.44 \cos(400t - 44.74^\circ).$$

Problem-19

Given $i_1(t) = 4 \cos(\omega t + 30^\circ) \text{ A}$ and $i_2(t) = 5 \sin(\omega t - 20^\circ) \text{ A}$, find their sum.

Solution:

$$i_1(t) = 4 \cos(\omega t + 30^\circ)$$

$$I_1 = 4 \angle 30^\circ \text{ A}$$

$$\begin{aligned} \text{And, } i_2(t) &= 5 \sin(\omega t - 20^\circ) \\ &= 5 \cos(\omega t - 20^\circ - 90^\circ) \\ &= 5 \cos(\omega t - 110^\circ) \\ &= 5 \angle -110^\circ. \end{aligned}$$

$$I = I_1 + I_2$$

$$= 4 \angle 30^\circ + 5 \angle -110^\circ.$$

$$= 3.46 + j2 - 1.71 - j4.7$$

$$= 1.75 - j2.7$$

$$= 3.218 \angle -57^\circ \text{ A}.$$

$$i(t) = 3.218 \cos(\omega t - 57^\circ) \text{ A}.$$

Problem-19(b)

Given $v_1 = -10 \sin(\omega t - 30^\circ) \text{ V}$ and $v_2 = 20 \cos(\omega t + 45^\circ) \text{ V}$, find

$$v = v_1 + v_2.$$

Solution:

$$v_1 = -10 \sin(\omega t - 30^\circ) \text{ V}$$
$$= 10 \cos(\omega t - 30^\circ + 90^\circ) \text{ V}$$

$$v_1 = 10 \cos(\omega t + 60^\circ) \text{ V}$$

$$V_1 = 10 \angle 60^\circ \text{ V.}$$

$$v_2 = 20 \cos(\omega t + 45^\circ)$$

$$V_2 = 20 \angle 45^\circ \text{ V}$$

$$V = V_1 + V_2 = 10 \angle 60^\circ + 20 \angle 45^\circ.$$

$$= 5 + j8.66 + 14.14 + j14.14$$

$$= 19.14 + j22.8$$

$$= 29.77 \angle 49.98^\circ$$

$$v(t) = 29.77 \cos(\omega t + 49.98^\circ) \text{ V.}$$

Problem-20

Two voltages v_1 and v_2 appear in series so that their sum

is $v = v_1 + v_2$. If $v_1 = 10 \cos(50t - \pi/3) \text{ V}$ and $v_2 = 12 \cos(50t + 30^\circ) \text{ V}$.

find v .

Solution:

$$v_1 = 10 \cos(50t - \pi/3) \text{ V}$$

$$= 10 \cos(50t - 60^\circ) \text{ V}$$

$$V_1 = 10 \angle -60^\circ \text{ V},$$

$$v_2 = 12 \cos(50t + 30^\circ) \text{ V}$$

$$V_2 = 12 \angle 30^\circ \text{ V}.$$

$$V = V_1 + V_2 = 10 \angle -60^\circ + 12 \angle 30^\circ$$

$$= 5 - j8.66 + 10.39 + j6,$$

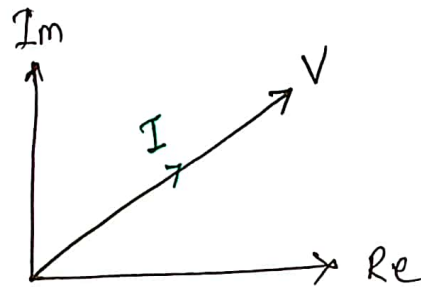
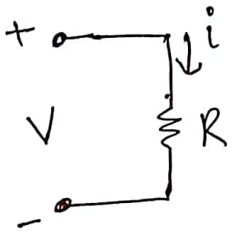
$$= 15.39 - j2.66$$

$$= 15.62 \angle -9.81^\circ.$$

$$v(t) = 15.62 \cos(50t - 9.81^\circ).$$

Phasor Relationship for Circuit Elements

Resistor

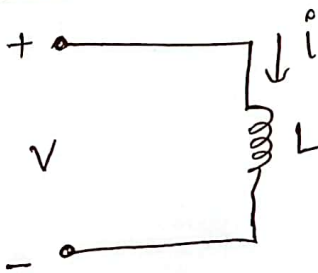


$$i = I_m \cos(\omega t + \phi)$$

$$I = I_m \angle \phi$$

$$\begin{aligned} V &= iR \\ &= I_m R \cos(\omega t + \phi) \\ &= V_m \cos(\omega t + \phi) \\ &= V_m \angle \phi. \end{aligned}$$

Inductor



$$i = I_m \cos(\omega t + \phi)$$

$$I = I_m \angle \phi.$$

$$V = L \cdot \frac{di}{dt}$$

$$= L \frac{d}{dt} (I_m \cos(\omega t + \phi))$$

$$= L I_m (-\sin(\omega t + \phi) \omega)$$

$$= -\omega L I_m \sin(\omega t + \phi)$$

$$= \omega L I_m \cos(\omega t + \phi + 90^\circ)$$

$$= \omega L I_m \angle \phi + 90^\circ.$$

The V-I relationship for Inductor:

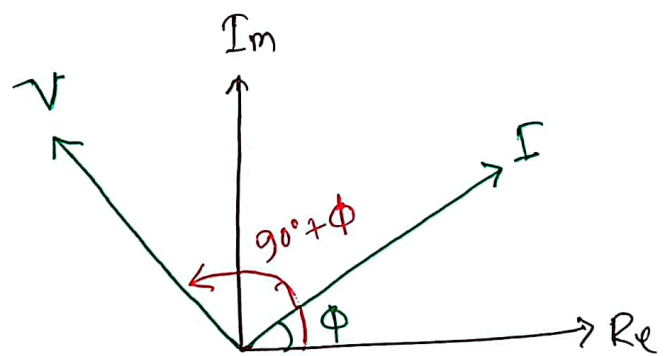
$$V = \omega L I_m \angle \phi + 90^\circ$$

$$= \omega L I \angle 90^\circ.$$

$$= \omega L I (\cos 90^\circ + j \sin 90^\circ)$$

$$= \omega L I j$$

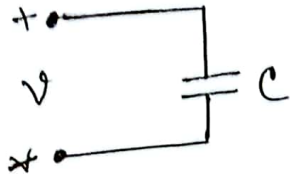
$$\boxed{V = j \omega L I}$$



V leads I by 90°

or, I lags V by 90° .

Capacitor:



$$i = I_m \cos(\omega t + \phi)$$

$$\Rightarrow I = I_m \angle \phi.$$

$$V(t) = \frac{1}{C} \int i dt$$

$$= \frac{1}{C} \int I_m \cos(\omega t + \phi) dt$$

$$= \frac{1}{C} I_m \frac{\sin(\omega t + \phi)}{\omega}.$$

$$= \frac{1}{\omega C} I_m \sin(\omega t + \phi)$$

$$= \frac{1}{\omega C} I_m \cos(\omega t + \phi - 90^\circ)$$

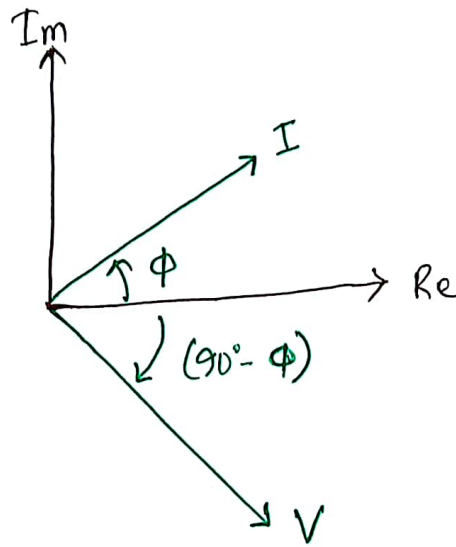
$$= \frac{1}{\omega C} I_m \angle \phi - 90^\circ.$$

$$= \frac{1}{\omega C} I \angle -90^\circ.$$

$$= \frac{I}{\omega C} [\cos(-90^\circ) + j \sin(-90^\circ)] \quad [\because r \angle \theta = r \cos \theta + j \sin \theta]$$

$$= - \frac{j}{\omega C} I$$

$$= \frac{I}{j \omega C}$$



I leads V by 90° or, V lags I by 90° .

The V - I relationship for a capacitor:

$$V = \frac{I}{j\omega C}$$

Comparison of Time Domain and Frequency-Domain Voltage-Current Expressions:

Time Domain

Frequency Domain

$$\begin{array}{c} i \\ \rightarrow \\ \text{---} R \text{---} \\ + \quad v \quad - \end{array} \quad v = iR$$

$$\begin{array}{c} I \\ \rightarrow \\ \text{---} R \text{---} \\ + \quad V \quad - \end{array} \quad V = RI$$

$$\begin{array}{c} i \\ \rightarrow \\ \text{---} L \text{---} \\ + \quad v \quad - \end{array} \quad v = L \cdot \frac{di}{dt}$$

$$\begin{array}{c} I \\ \rightarrow \\ \text{---} j\omega L \text{---} \\ + \quad V \quad - \end{array} \quad V = j\omega L I$$

$$\begin{array}{c} i \\ \rightarrow \\ \text{---} \frac{1}{C} \text{---} \\ + \quad v \quad - \end{array} \quad v = \frac{1}{C} \int i dt$$

$$\begin{array}{c} I \\ \rightarrow \\ \text{---} \frac{1}{j\omega C} \text{---} \\ + \quad V \quad - \end{array} \quad V = \frac{1}{j\omega C} I$$

Impedance, Reactance & Admittance

We obtained the voltage current relations for the three passive elements as

$$V = RI \quad ; \quad V = j\omega LI \quad ; \quad V = \frac{1}{j\omega C} I.$$

These equations may be written in terms of the ratio of the phasor voltage to the phasor current as

$$\frac{V}{I} = R \quad ; \quad \frac{V}{I} = j\omega L \quad ; \quad \frac{V}{I} = \frac{1}{j\omega C}.$$

We obtain Ohm's law in phasor form for any type of element as

$$Z = \frac{V}{I} \quad \text{or,} \quad V = ZI.$$

where Z is a frequency-dependent quantity known as impedance.

The impedance represents the opposition that the circuit exhibits to the flow of sinusoidal current.

As a complex quantity, the impedance may be expressed as

$$Z = R + jX$$

where, X is the reactance.

The reactance X may be positive or negative.

Reactance:

The imaginary part of the impedance is called reactance.

Inductive reactance: $X_L = \omega L$

Capacitive reactance: $X_C = -\frac{1}{\omega C}$.

The impedance may be expressed as polar form

$$Z = |Z| \angle \theta.$$

where,

$$Z = \sqrt{R^2 + X^2} \quad ; \quad \theta = \tan^{-1} \frac{X}{R}.$$

$$R = |Z| \cos \theta \quad ; \quad X = |Z| \sin \theta.$$

Admittance

The admittance, Y is the reciprocal of impedance, measured in Siemens (S). Admittance is a measure of how easily a circuit can conduct AC current.

$$Y = \frac{1}{Z}.$$

Impedance, Reactance and Admittances of Passive elements

<u>Element</u>	<u>Impedance</u>	<u>Admittance</u>	<u>Reactance</u>
R	$Z = R$	$Y = \frac{1}{R}$	-
L	$Z_L = j\omega L$	$Y = \frac{1}{j\omega L}$	$X_L = \omega L$
C	$Z_C = \frac{1}{j\omega C} = -\frac{j}{\omega C}$	$Y = j\omega C$	$X_C = -\frac{1}{\omega C}$

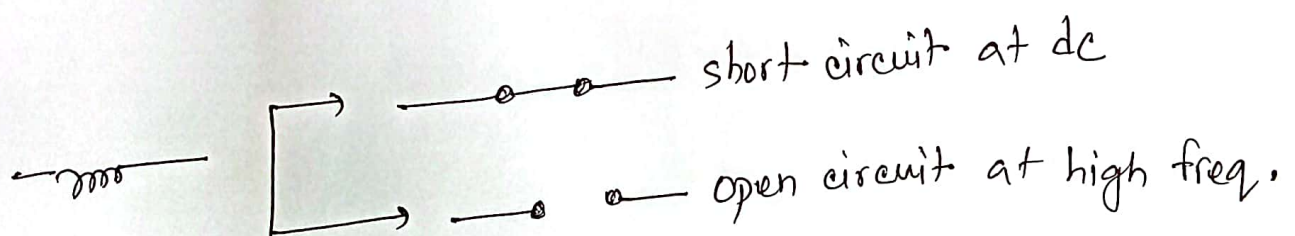
Inductor and capacitor for dc and ac supply

Inductor: $Z_L = j\omega L = j2\pi f L$.

For dc, $f=0$, $Z_L=0$.

For ac, f is high; Z_L is high; $V_L = I_L Z_L \propto [f \uparrow]$

$$\Rightarrow I_L = 0.$$

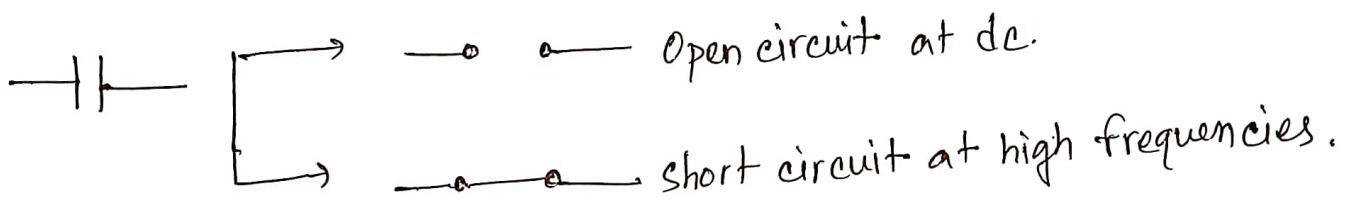


Capacitor:

$$Z_c = \frac{1}{j\omega C} = \frac{1}{j 2\pi f C}$$

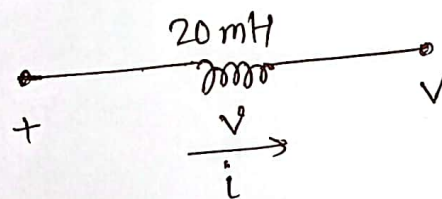
At dc, $f=0$, $Z_c \approx \infty$; open circuit

At ac, $f \uparrow \uparrow$, $Z_c \downarrow \downarrow$; $V_c = I_c Z_c \approx 0V$



Problem - 21

The current in the 20 mH inductor is $10 \cos(10,000t + 30^\circ)$ mA. Calculate (a) the inductive reactance; (b) the impedance of the inductor; (c) the phasor voltage, V ; and (d) the steady-state expression for $v(t)$.



Solution: $L = 20 \text{ mH} = 20 \times 10^{-3} \text{ H}$

$$i(t) = 10 \cos(10,000t + 30^\circ) \text{ mA} \\ = 10 \times 10^{-3} \cos(10,000t + 30^\circ)$$

where, $\omega = 10,000 \text{ rad s}^{-1}$.

(a) the inductive reactance, $X_L = \omega L$

$$= 10,000 \times 20 \times 10^{-3} \Omega.$$

$$= 200 \Omega.$$

(b) the impedance of the inductor, $Z_L = j\omega L = j200 \Omega.$

(c) the phasor voltage, $V_L = I Z_L$

$$= 10 \times 10^{-3} \cos(10,000t + 30^\circ) \times j200.$$

$$= (10 \times 10^{-3} \angle 30^\circ) \times (200 \angle 90^\circ)$$

$$= 2 \angle 120^\circ \text{ V}.$$

$$(d) v_L(t) = 2 \cos(10,000t + 120^\circ) \text{ V}.$$

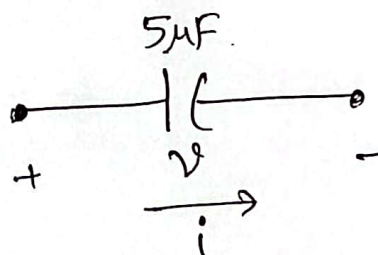
Problem-22

The voltage across the terminals of the $5 \mu\text{F}$ capacitor is

$30 \cos(4000t + 25^\circ) \text{ V}$. Calculate (a) the capacitive reactance;

(b) the impedance of the capacitor; (c) the phasor current, I ;

and (d) the steady-state expression for $i(t)$.



Solution:

$$v(t) = 30 \cos(4000t + 25^\circ) \text{ V}$$

$$\omega = 4000 \text{ rad s}^{-1}.$$

$$C = 5 \mu\text{F} = 5 \times 10^{-6} \text{ F}.$$

(a) the capacitive reactance, $X_C = -\frac{1}{\omega C}$.

$$= -\frac{1}{4000 \times 5 \times 10^{-6}}$$
$$= -50 \Omega.$$

(b) the impedance of the capacitor, $Z_L = jX_C = -j50 \Omega$.

(c) the phasor current, $I = \frac{V}{Z_L} = \frac{30 \angle 25^\circ}{-j50}$

$$= \frac{30 \angle 25^\circ}{50 \angle -90^\circ}$$

$$= 0.6 \angle 115^\circ \text{ A}.$$

(d) the steady-state expression for $i(t)$

$$I(t) = 0.6 \angle 115^\circ \text{ A}$$

$$i(t) = 0.6 \cos(4000t + 115^\circ) \text{ A}.$$

Problem-23

A 80 kHz sinusoidal voltage has zero phase angle and a maximum amplitude of 25 mV. When this voltage is applied across the terminals of a capacitor, the resulting steady-state current has a maximum amplitude of 628.32 μA .

- What is the frequency of the current in radians per second?
- What is the phase angle of the current?
- What is the capacitive reactance of the capacitor?
- What is the capacitance of the capacitor in microfarads?
- What is the impedance of the capacitor?

Solution:

$$a) \omega = 2\pi f = 2\pi \times 80 \times 10^3 \text{ rad s}^{-1} = 502.656 \text{ rad s}^{-1} = 502.656 \text{ krad s}^{-1}.$$

$$b) I = \frac{V}{Z_c} = \frac{V}{\frac{1}{j\omega C}} = \frac{V}{-j \frac{1}{\omega C}} = \omega C \frac{V}{-j} = \omega C \frac{25 \times 10^{-3} \angle 0^\circ}{1 \angle -90^\circ}.$$

$$\Rightarrow I = 25 \times 10^{-3} \omega C \angle 90^\circ.$$

$$\therefore \theta_i = 90^\circ.$$

$$c) 628.32 \times 10^{-6} = 25 \times 10^{-3} \omega C$$

$$\Rightarrow \frac{1}{\omega C} = \frac{25 \times 10^{-3}}{628.32 \times 10^{-6}} = 39.79 \Omega$$

$$\therefore X_c = -\frac{1}{\omega C} = -39.79 \Omega$$

(d) The capacitance

$$\frac{1}{\omega C} = 39.79$$

$$\Rightarrow C = \frac{1}{\omega \times 39.79} = \frac{1}{502656 \times 39.79} = 5 \times 10^{-8} \text{ F} = 0.05 \mu\text{F}$$

$$\therefore C = 0.05 \mu\text{F}.$$

(e) the impedance of the capacitor, $Z_C = jX_C = -j 39.79 \Omega$.

Problem - 24

A 400 Hz sinusoidal voltage with a maximum amplitude of 100V at $t=0$ is applied across the terminals of an inductor. The maximum amplitude of the steady-state current in the inductor is 20A.

- What is the frequency of the inductor current?
- If the phase angle of the voltage is zero, what is the phase angle of the current?
- What is the inductive reactance of the inductor?
- What is the inductance of the inductor in millihenrys?
- What is the impedance of the inductor?

Solution:

a) 400 Hz

b) $\theta_v = 0$

$$I = \frac{V}{Z_L} = \frac{100 \angle 0^\circ}{j\omega L} = \frac{100 \angle 0^\circ}{\omega L \angle 90^\circ} = \frac{100}{\omega L} \angle -90^\circ.$$

$$\theta_i = -90^\circ.$$

c) the inductive reactance,

$$X_L = \omega L$$

$$I_m = 20 \text{ A}$$

$$\Rightarrow \frac{100}{\omega L} = 20 \Rightarrow \omega L = \frac{100}{20} = 5 \Omega$$

$$\therefore X_L = 5 \Omega.$$

d) $\omega L = 5 \Omega$

$$\Rightarrow L = \frac{5}{\omega} = \frac{5}{2\pi f} = \frac{5}{2\pi \times 400} = 1.989 \times 10^{-3} \text{ H}$$
$$= 1.989 \text{ mH}.$$

e) $Z_L = j\omega L = j5 \Omega.$

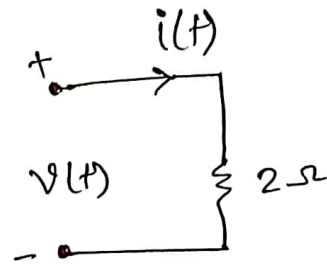
Problem-25

Calculate the current in the resistor in Fig. if the voltage input is

a) $V_1(t) = 10 \cos(377t + 180^\circ) \text{ V}$

b) $V_2(t) = 12 \sin(377t + 45^\circ) \text{ V}$

Give the answers in both the time and frequency domains.



Solution:

$$a) i_1(t) = \frac{V_1(t)}{R} = \frac{10 \cos(377t + 180^\circ)}{2} = 5 \cos(377t + 180^\circ)$$

$$I_1 = 5 \angle 180^\circ \text{ A}$$

$$b) i_2(t) = \frac{V_2(t)}{R} = \frac{12 \sin(377t + 45^\circ)}{2}$$

$$= 6 \sin(377t + 45^\circ)$$

$$= 6 \cos(377t + 45^\circ - 90^\circ)$$

$$= 6 \cos(377t - 45^\circ)$$

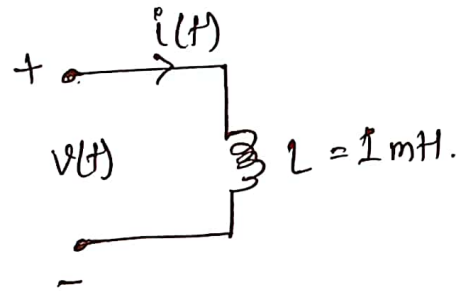
$$I_2 = 6 \angle -45^\circ \text{ A}.$$

Problem-26

Calculate the current in the inductor shown in Fig. if the voltage input is

a) $v_1(t) = 10 \cos(377t + 45^\circ) \text{ V}$

b) $v_2(t) = 5 \sin(377t - 90^\circ) \text{ V}$



Give the answers in both the time and frequency domains.

Solution:

$$a) i_1(t) = \frac{1}{L} \int v_1(t) dt = \frac{1}{1 \times 10^{-3}} \int 10 \cos(377t + 45^\circ) dt$$

$$= \frac{1}{1 \times 10^{-3}} \times 10 \left[\sin(377t + 45^\circ) \cdot \frac{1}{377} \right]$$

$$= \frac{10}{1 \times 10^{-3} \times 377} \sin(377t + 45^\circ)$$

$$= 26.53 \sin(377t + 45^\circ)$$

$$= 26.53 \cos(377t + 45^\circ - 90^\circ)$$

$$= 26.53 \cos(377t - 45^\circ)$$

$$I_1 = 26.53 \angle -45^\circ \text{ A.}$$

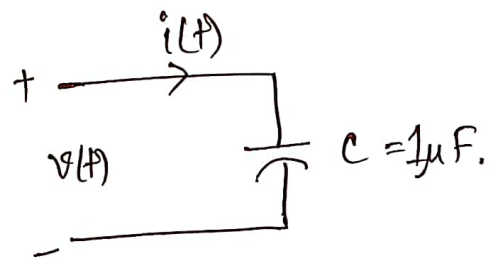
$$\begin{aligned}
 b) \ i_2(t) &= \frac{1}{L} \int v_2(t) dt \\
 &= \frac{1}{L} \int 5 \sin(377t - 90^\circ) dt \\
 &= \frac{5}{L} \left[\frac{-\cos(377t - 90^\circ)}{377} \right] \\
 &= \frac{5}{377 \times 10^{-3}} [-\cos(377t - 90^\circ)] \\
 &= -13.26 \cos(377t - 90^\circ) \\
 &= 13.26 \cos(377t - 90^\circ + 180^\circ) \\
 &= 13.26 \cos(377t + 90^\circ) \\
 I_2 &= 13.26 \angle 90^\circ \text{ A.}
 \end{aligned}$$

Problem - 27

Calculate the current in the capacitor shown in Fig. if the voltage input is

$$a) \ v_1(t) = 10 \cos(377t - 30^\circ) \text{ V}$$

$$b) \ v_2(t) = 5 \sin(377t + 60^\circ) \text{ V}$$



Give the answers in both the time and frequency domains.

Solution:

$$a) i_1(t) = C \frac{dv_1(t)}{dt} = 10^{-6} \times \frac{d}{dt} (10 \cos(377t - 30^\circ))$$

$$= 10^{-6} \times 10 [-\sin(377t - 30^\circ) \times 377]$$

$$= -0.00377 \sin(377t - 30^\circ)$$

$$= 0.00377 \cos(377t - 30^\circ + 90^\circ)$$

$$\dot{i}_1(t) = 0.00377 \cos(377t + 60^\circ)$$

$$I_1 = 0.00377 \angle 60^\circ.$$

$$b) i_2(t) = C \frac{dv_2(t)}{dt}$$

$$= 10^{-6} \times \frac{d}{dt} [5 \sin(377t + 60^\circ)]$$

$$= 10^{-6} \times 5 [\cos(377t + 60^\circ)] \times 377$$

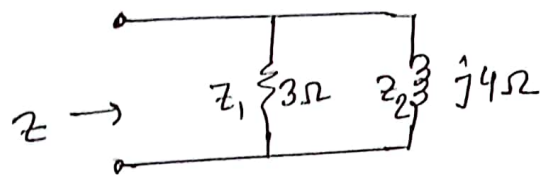
$$= 10^{-6} \times 5 \times 377 \cos(377t + 60^\circ)$$

$$= 0.001885 \cos(377t + 60^\circ)$$

$$I_2 = 0.001885 \angle 60^\circ.$$

Problem - 28

Find the frequency-domain impedance, Z , as shown in Fig.



Solution:

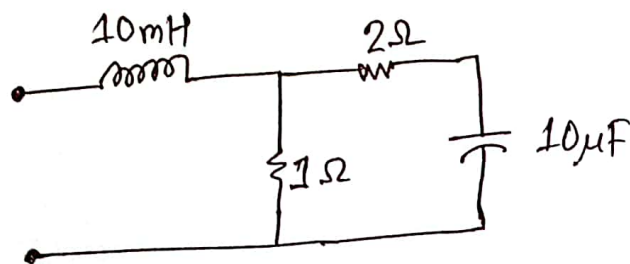
$$Z = \left(\frac{1}{Z_1} + \frac{1}{Z_2} \right)^{-1} = \left(\frac{1}{3} + \frac{1}{j4} \right)^{-1}$$

$$= 1.92 + j1.44$$

$$Z = 2.4 \angle 36.87^\circ \Omega$$

Problem - 29

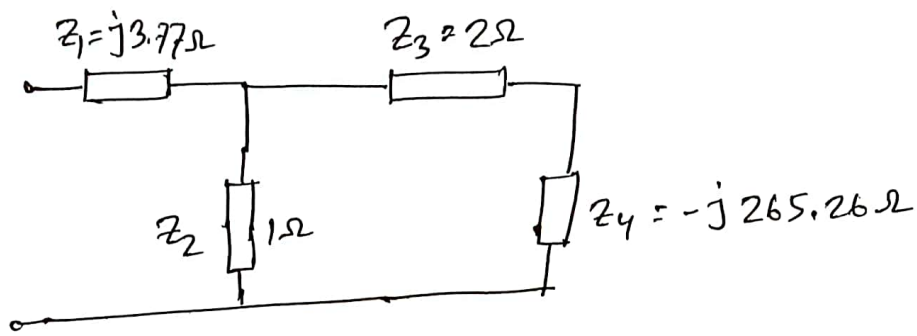
Find the impedance, Z , shown in Fig. at a frequency of 60 Hz.



Solution:

$$\begin{aligned} \text{The impedance of } 10\text{mH} &= j\omega L = j2\pi f L \\ &= j \times 2\pi \times 60 \times 10 \times 10^{-3} \Omega \\ &= j3.77 \Omega \end{aligned}$$

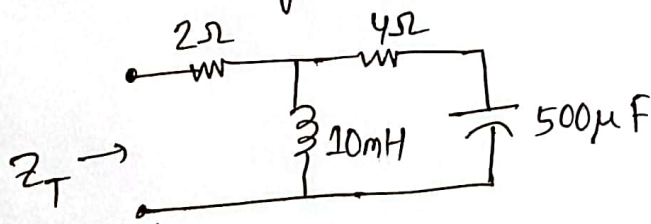
$$\begin{aligned} \text{The impedance of } 10\mu\text{F} &= \frac{1}{j\omega C} = \frac{1}{j2\pi f C} = \frac{1}{j \times 2\pi \times 60 \times 10 \times 10^{-6}} \Omega \\ &= -j265.26 \Omega \end{aligned}$$



$$\begin{aligned}
 Z_{eq} &= (Z_3 + Z_4) \parallel Z_2 + Z_1 \\
 &= (2 - j265.26 \parallel 1) + j3.77 \\
 &= \left(\frac{1}{2 - j265.26} + \frac{1}{1} \right)^{-1} + j3.77 \\
 &= 1 - j0.00377 + j3.77 \\
 &= 1 + j3.766 \Omega.
 \end{aligned}$$

Problem-30

In the network in Fig., find Z at a frequency of 60 Hz.

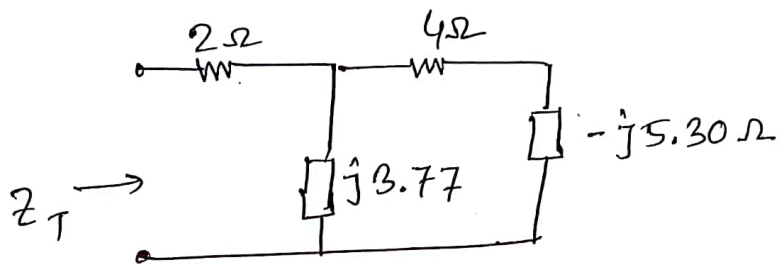


Solution:

$$\omega = 2\pi f = 2\pi \times 60 = 377 \text{ rad s}^{-1}.$$

$$\begin{aligned}
 \text{The impedance of } 10\text{mH} &= j\omega L = j377 \times 10 \times 10^{-3} \Omega \\
 &= j3.77 \Omega
 \end{aligned}$$

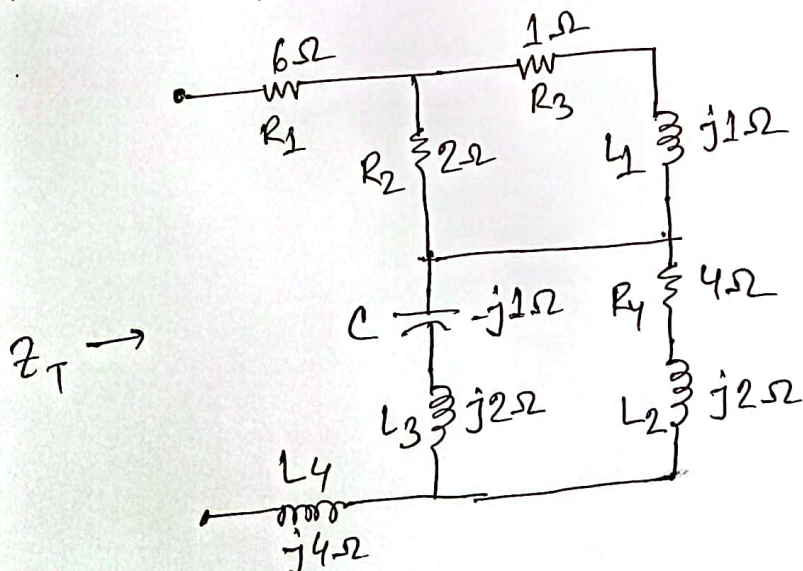
$$\text{The impedance of } 500\mu\text{F} = \frac{1}{j\omega C} = \frac{1}{j \times 377 \times 500 \times 10^{-6}} = -j5.30 \Omega.$$



$$\begin{aligned}
 z_T &= ((4 - j5.30) \parallel j3.77) + 2 \\
 &= \left(\frac{1}{4 - j5.30} + \frac{1}{j3.77} \right)^{-1} + 2 \\
 &= 3.1 + j4.95 + 2 \\
 &= 5.1 + j4.95 \Omega \\
 &= 7.11 \angle 44.17^\circ
 \end{aligned}$$

Problem-31

Find the frequency-domain impedance, z .



Set

Solution:

R_3 and L_1 are in series, $z_1 = 1 + j1 \Omega$

R_4 and L_2 are in series, $z_2 = 4 + j2 \Omega$

C and L_3 are in series, $z_3 = j2 - j1 = j1 \Omega$

$$z_1 \text{ and } R_2 \text{ are in parallel, } z_4 = \left(\frac{1}{z_1} + \frac{1}{R_2} \right)^{-1} \\ = 0.8 + j0.4 \Omega$$

$$z_2 \text{ and } z_3 \text{ are in parallel, } z_5 = \left(\frac{1}{4+j2} + \frac{1}{j1} \right)^{-1} \\ = 0.16 + j0.88 \Omega$$

R_1, z_4, z_5 and L_4 are in series.

Hence, the total impedance,

$$z_T = R_1 + z_4 + z_5 + L_4$$

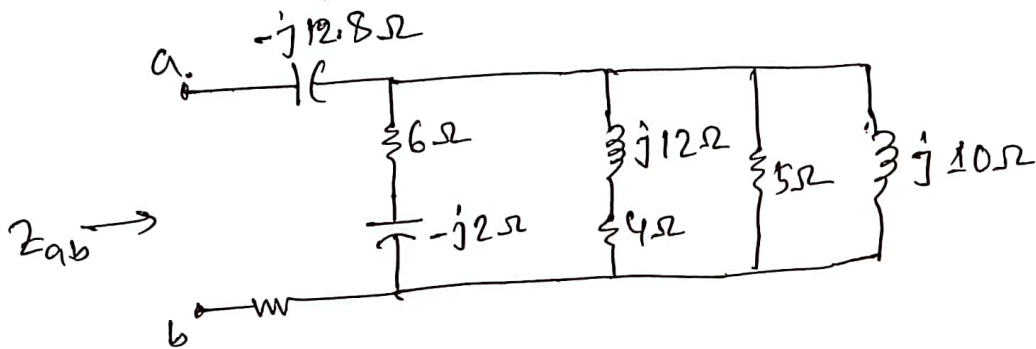
$$= 6 + 0.8 + j0.4 + 0.16 + j0.88 + j4$$

$$= 6.96 + j5.28 \Omega$$

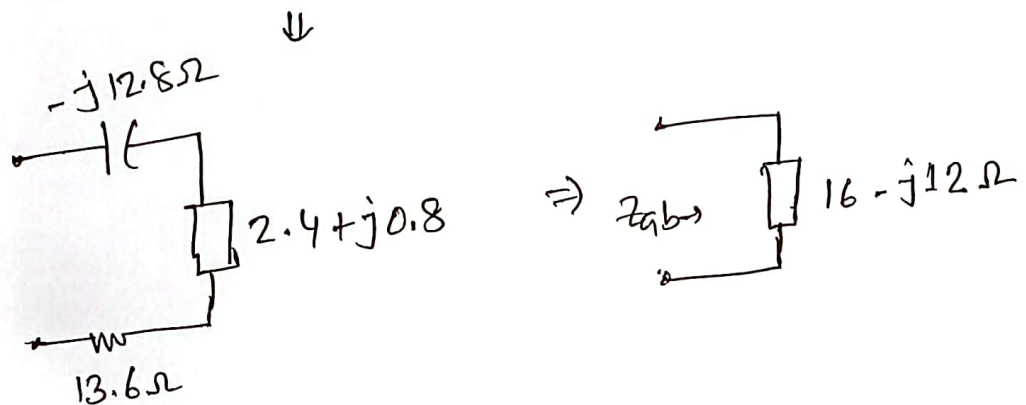
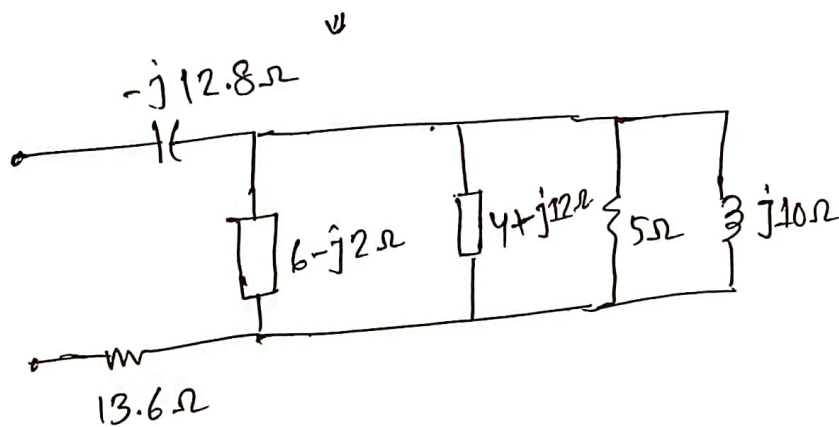
$$= 8.74 \angle 37.18^\circ.$$

Problem - 32

Find the impedance, Z_{ab} in the circuit in Fig. Express Z_{ab} in both polar and rectangular form.



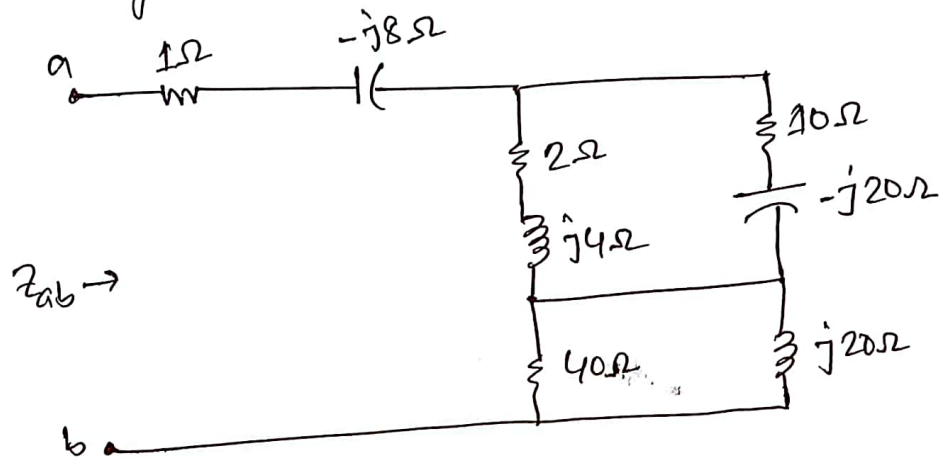
Solution:



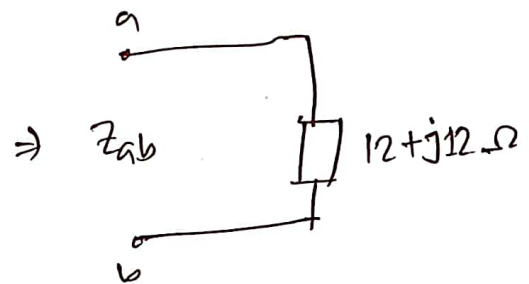
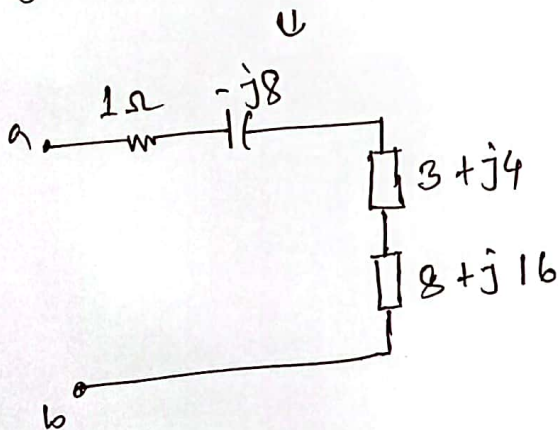
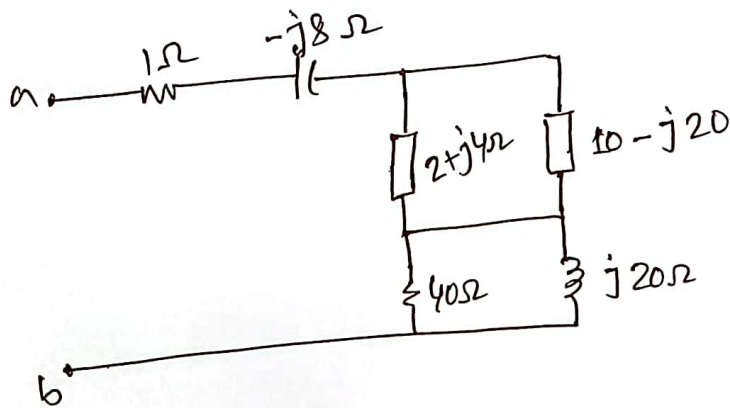
$$\begin{aligned} Z_{ab} &= 16 - j12 \Omega \\ &= 20 \angle -36.87^\circ \Omega. \end{aligned}$$

Problem-33

Find the impedance, Z_{ab} in the circuit in Fig. Express Z_{ab} in both polar and rectangular form.



Solution:



$$Z_{ab} = 12 + j12 \, \Omega = 16.97 \angle 45^\circ \, \Omega.$$

Problem - 34

Four branches terminate at a common node. The reference direction of each branch current (i_1, i_2, i_3 and i_4) is toward the node. If

$$i_1 = 100 \cos(\omega t + 25^\circ) \text{ A}$$

$$i_2 = 100 \cos(\omega t + 145^\circ) \text{ A, and}$$

$$i_3 = 100 \cos(\omega t - 95^\circ) \text{ A, find}$$

Solution:

$$I_1 = 100 \angle 25^\circ \text{ A}$$

$$I_2 = 100 \angle 145^\circ \text{ A}$$

$$I_3 = 100 \angle -95^\circ \text{ A}$$

$$I_1 + I_2 + I_3 + I_4 \Rightarrow I_4 = -(I_1 + I_2 + I_3)$$

$$\Rightarrow I_4 = -(100 \angle 25^\circ + 100 \angle 145^\circ + 100 \angle -95^\circ)$$

$$= -(90.63 + j42.26 - 81.91 + j57.36 - 8.71 - j99.62)$$

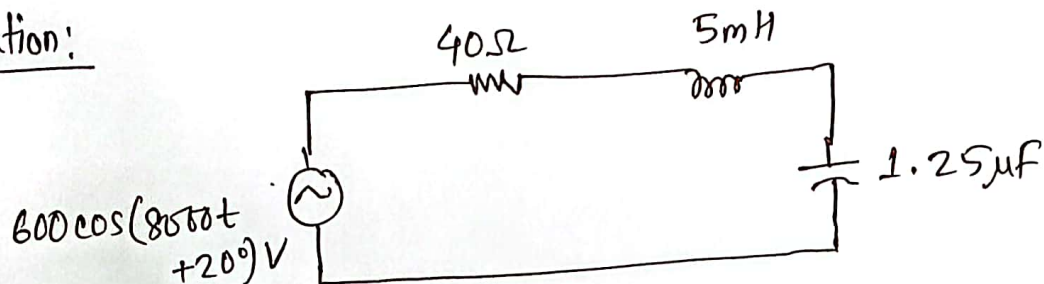
$$= 0 \text{ A.}$$

Problem-35

A 40Ω resistor, a 5mH inductor, and a $1.25\mu\text{F}$ capacitor are connected in series. The series-connected elements are energized by a sinusoidal voltage source whose voltage is $600\cos(8000t + 20^\circ)\text{V}$.

- Draw the frequency-domain equivalent circuit.
- Reference the current in the direction of the voltage rise across the source, and find the phasor current.
- Find the steady-state expression for $i(t)$.

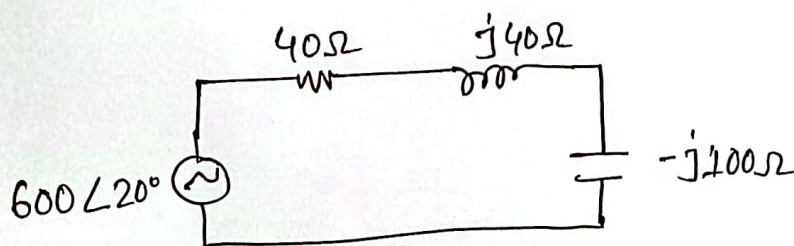
Solution:



$$a) \quad z_L = j\omega L = j8000 \times 5 \times 10^{-3} = j40\Omega$$

$$z_C = \frac{1}{j\omega C} = -j \cdot \frac{1}{8000 \times 1.25 \times 10^{-6}} = -j100\Omega$$

$$v(t) = 600\cos(8000t + 20^\circ)\text{V} = 600\angle 20^\circ\text{V}$$



$$(b) I = \frac{V}{Z_T} = \frac{600 \angle 20^\circ}{40 + j40 - j100} = 8.321 \angle 76.31^\circ A.$$

$$c) I = 8.321 \angle 76.31^\circ A$$

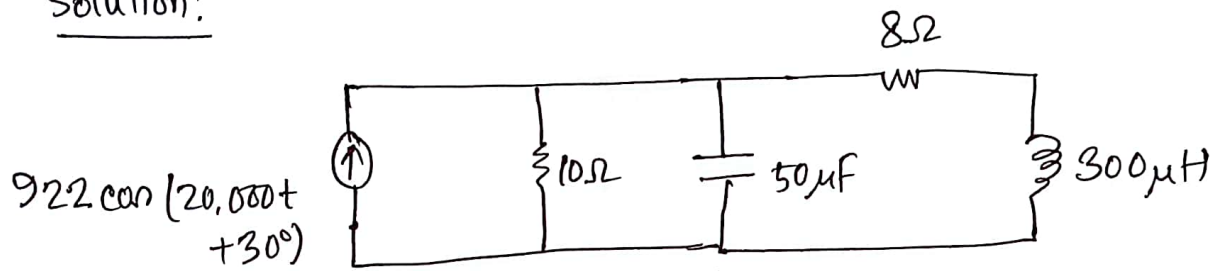
$$i(t) = 8.321 \cos(8000t + 76.31^\circ) A.$$

Problem-36

A 10Ω resistor and a $5\mu F$ capacitor are connected in parallel. This parallel combination is also in parallel with the series combination of an 8Ω resistor and a $300\mu H$ inductor. These parallel branches are driven by a sinusoidal current source where current is $922 \cos(20,000t + 30^\circ) A$.

- Draw the frequency domain equivalent circuit.
- Reference the voltage across the current source as a rise in the direction of the source current, and find the phasor voltage.
- Find the steady-state expression for $v(t)$.

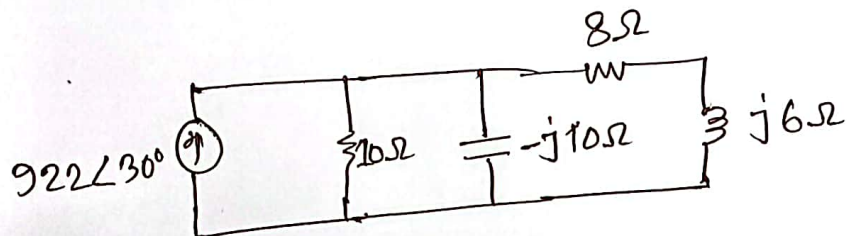
Solution:



a) $I = 922 \cos(20,000t + 30^\circ) \text{ A} = 922 \angle 30^\circ, \omega = 20,000 \text{ rad/s}.$

$$Z_L = j\omega L = j(20,000 \times 300 \times 10^{-6}) = j6 \Omega.$$

$$Z_C = \frac{1}{j\omega C} = \frac{1}{j(20,000 \times 5 \times 10^{-6})} = -j10 \Omega.$$



b) $Z_T = \left(\frac{1}{10} + \frac{1}{-j10} + \frac{1}{8+j6} \right)^{-1} = (5.294 - j1.176) \Omega = 5.42 \angle -12.53^\circ.$

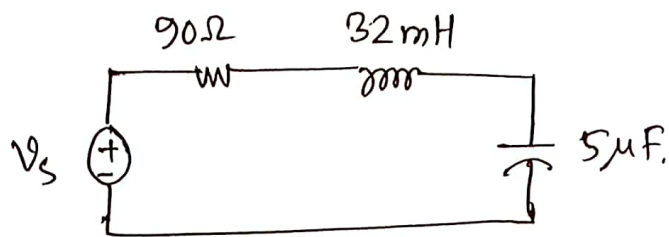
$$I = 922 \angle 30^\circ.$$

$$V_o = I Z_T = (922 \angle 30^\circ) (5.42 \angle -12.53^\circ) = 4997.24 \angle 17.47^\circ.$$

c) $v(t) = 4997.24 \cos(20,000t + 17.47^\circ) \text{ V}.$

Problem-37

A 90Ω resistor, a 32mH inductor, and a $5\mu\text{F}$ capacitor are connected in series across the terminals of a sinusoidal voltage source, as shown in Fig. The steady-state expression for the source voltage, v_s is $750 \cos(5000t + 30^\circ)\text{V}$.



- Construct the frequency-domain equivalent circuit.
- Calculate the steady-state current

Solution:

$$v_s = 750 \cos(5000t + 30^\circ)\text{V}$$

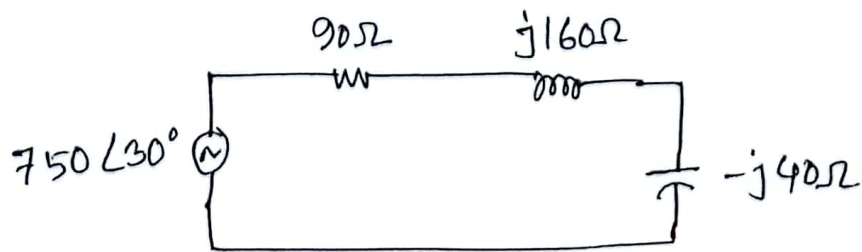
$$\vec{V}_s = 750 \angle 30^\circ \text{V}, \quad \omega = 5000 \text{ rad/s}.$$

The impedance of 32mH inductor is, $Z_L = j\omega L$

$$= j(5000 \times 32 \times 10^{-3})\Omega$$
$$= j160\Omega.$$

The impedance of $5\mu\text{F}$ capacitor, $Z_C = -j \cdot \frac{1}{\omega C} = -j \frac{1}{5000 \times 5 \times 10^{-6}}$

$$= -j40\Omega.$$



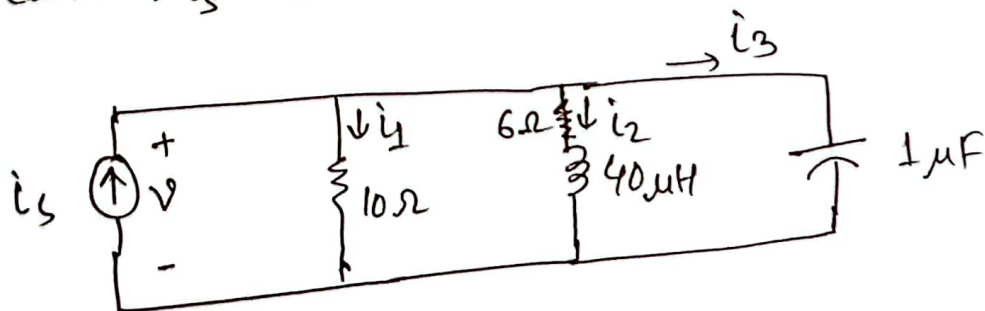
$$b) Z_T = 90 + j160 - j40 = 90 + j120 \\ = 150 \angle 53.13^\circ \Omega$$

$$\text{Thus, } I = \frac{V}{Z_T} = \frac{750 \angle 30^\circ}{150 \angle 53.13^\circ} = 5 \angle -23.13^\circ \text{ A}$$

$$i(t) = 5 \cos(5000t - 23.13^\circ) \text{ A}$$

Problem-38

The sinusoidal current source in the circuit shown in Fig. produces the current, $i_s = 8 \cos 200,000t \text{ A}$.



- Construct the frequency-domain equivalent circuit.
- Find the steady-state expressions for v , i_1 , i_2 , and i_3 .

Solution:

The phasor of current source, $i_s = 8 \cos 200,000t$ A

$$I_s = 8 \angle 0^\circ, \omega = 200,000 \text{ rad s}^{-1}.$$

The impedance of $40\mu\text{H}$, $Z_L = j\omega L = j \times 200,000 \times 40 \times 10^{-6} \Omega$
 $= j8 \Omega.$

The impedance of $1\mu\text{C}$, $Z_C = \frac{1}{j\omega C} = \frac{1}{j \times 200,000 \times 10^{-6}} \Omega$
 $= -j5 \Omega.$

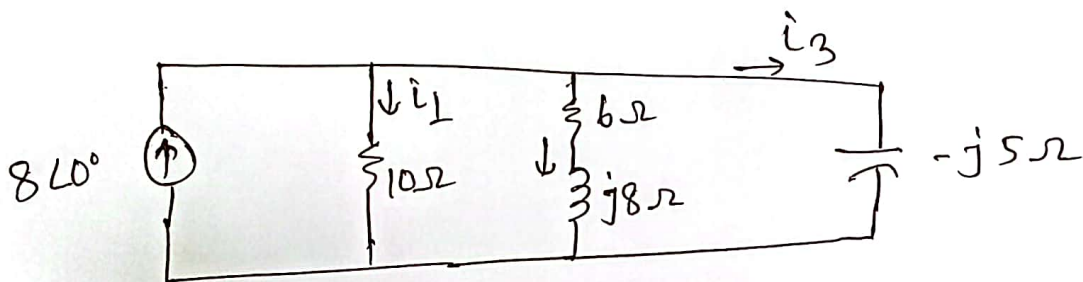


Fig. The frequency-domain equivalent circuit.

b) The total impedance

$$Z_T = \left(\frac{1}{10} + \frac{1}{6+j8} + \frac{1}{-j5} \right)^{-1} = 4 - j3 \Omega = 5 \angle -36.87^\circ \Omega.$$

The voltage, $V = IZ$

$$= (8 \angle 0^\circ)(5 \angle -36.87^\circ)$$

$$= 40 \angle -36.87^\circ$$

Hence, $v = 40 \cos(200,000t - 36.87^\circ) \text{ V}.$

$$I_1 = \frac{V}{10} = \frac{40 \angle -36.87^\circ}{10} = 4 \angle -36.87^\circ$$

$$i_1(t) = 4 \cos(200,000t - 36.87^\circ)$$

$$I_2 = \frac{V}{6+j8} = \frac{40 \angle -36.87^\circ}{10 \angle 53.13^\circ} = 4 \angle -90^\circ$$

$$i_2(t) = 4 \cos(200,000t - 90^\circ) \text{ A}$$

$$I_3 = \frac{V}{-j5} = \frac{40 \angle -36.87^\circ}{5 \angle -90^\circ} = 8 \angle 53.13^\circ$$

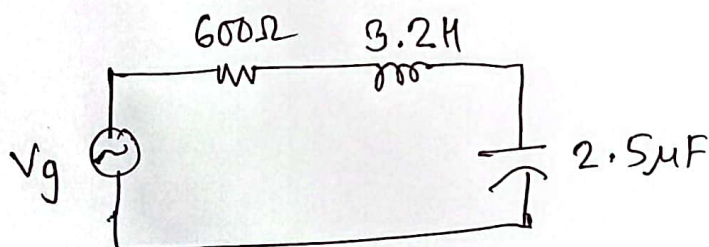
$$i_3(t) = 8 \cos(200,000t + 53.13^\circ)$$

Problem - 37

The circuit shown in Fig. is operating in the sinusoidal steady state. Find the value of ω if

$$i_o = 40 \sin(\omega t + 21.87^\circ) \text{ mA}$$

$$v_g = 40 \cos(\omega t - 15^\circ) \text{ V,}$$



Solution:

$$i_o = 40 \sin(\omega t + 21.87^\circ) \text{ mA}$$

$$= 40 \times 10^{-3} \cos(\omega t + 21.87^\circ - 90^\circ) \text{ A}$$

$$= 40 \times 10^{-3} \cos(\omega t - 68.13^\circ) \text{ A}$$

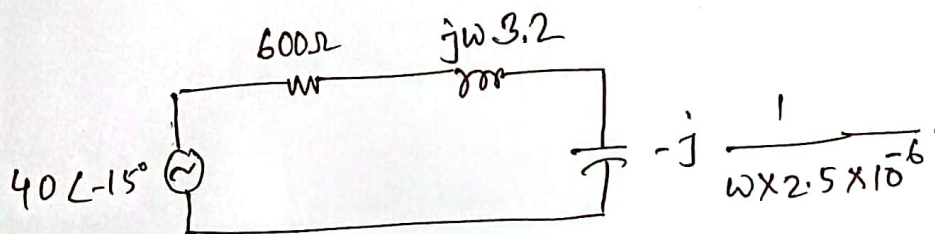
$$I_o = 40 \times 10^{-3} \angle -68.13^\circ \text{ A}$$

$$V_g = 40 \cos(\omega t - 15^\circ) \text{ V}$$

$$V_g = 40 \angle -15^\circ \text{ V}$$

$$Z = \frac{V_g}{I_o} = \frac{40 \angle -15^\circ}{40 \times 10^{-3} \angle -68.13^\circ} = 1000 \angle 53.13^\circ \Omega$$
$$= 600 + j800 \Omega \quad \text{--- (1)}$$

The frequency-domain equivalent circuit



$$\text{So, } Z_T = 600 + j\omega 3.2 - j \frac{1}{\omega \times 2.5 \times 10^{-6}}$$

$$= 600 + j \left(3.2\omega - \frac{1}{\omega \times 2.5 \times 10^{-6}} \right) \quad \text{--- (11)}$$

By comparing equⁿ ① and ②

$$3.2\omega - \frac{1}{\omega \times 2.5 \times 10^{-6}} = 800$$

$$\Rightarrow 3.2\omega - \frac{4 \times 10^5}{\omega} = 800$$

$$\Rightarrow 3.2\omega^2 - 4 \times 10^5 = 800\omega$$

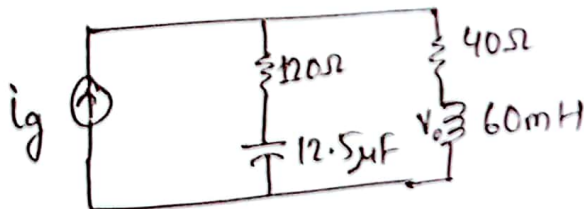
$$\Rightarrow 3.2\omega^2 - 800\omega - 4 \times 10^5 = 0$$

$$\omega = 500 \text{ rad s}^{-1}.$$

Problem - 40

Find the steady-state expression for v_o in the circuit of fig. if

$$i_g = 500 \cos 2000t \text{ mA}.$$



Solution:

The impedance of $12.5\mu\text{F}$ capacitor, $Z_c = \frac{1}{j\omega C}$

$$= \frac{1}{j 2000 \times 12.5 \times 10^{-6}}$$

$$= -j40\Omega.$$

The impedance of 60 mH inductor, $z_L = j\omega L$

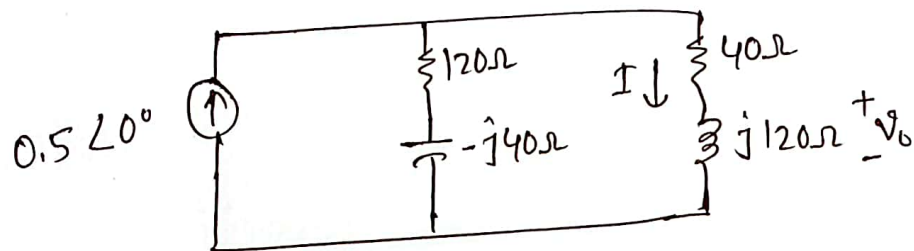
$$= j \times 2000 \times 60 \times 10^{-3}$$

$$= j120\Omega$$

$$i_g = 500 \cos 2000t \text{ mA}$$

$$I_g = 500 \times 10^{-3} \angle 0^\circ = 0.5 \angle 0^\circ$$

Construct the phasor domain equivalent circuit.



Using current division: $I = \frac{z_T}{z_x} I_g$

$$z_T = \left(\frac{1}{120 - j40} + \frac{1}{40 - j120} \right)^{-1} = 80 + j40\Omega$$

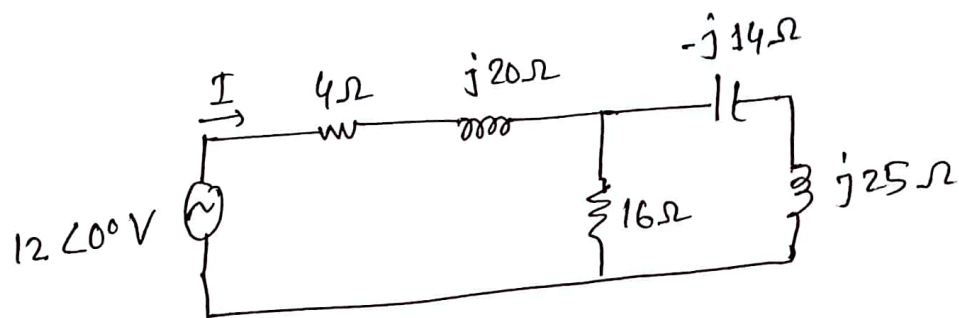
$$I = \frac{80 + j40}{40 + j120} \times 0.5 \angle 0^\circ = 0.354 \angle -44.99^\circ$$

$$\begin{aligned} V_o &= j120 I = (j120) (0.354 \angle -44.99^\circ) \\ &= (120 \angle 90^\circ) (0.354 \angle -44.99^\circ) \\ &= 42.43 \angle 45^\circ \end{aligned}$$

$$v_o(t) = 42.43 \cos(2000t + 45^\circ) \text{ V}$$

Problem - 41

Find the circuit shown in fig, find Z_{eq} and use that to find current I . Let $\omega = 10 \text{ rad/s}$.



Solution:

$$Z_{eq} = \cancel{(-j14)}$$

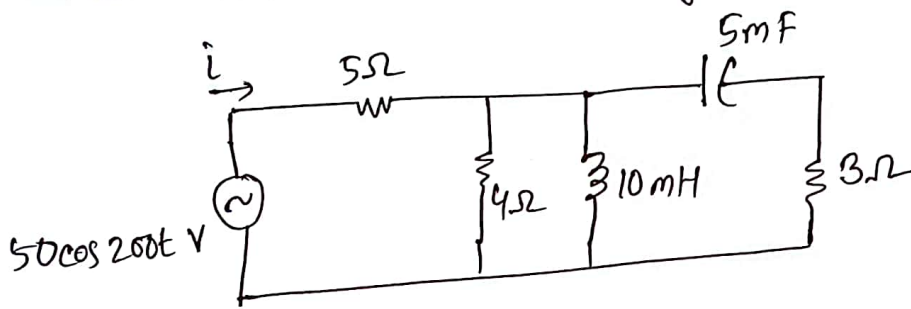
$$\begin{aligned} Z_{eq} &= (-j14 + j25) \parallel 16 + j20 + 4 \\ &= 5.135 + j7.47 + j20 + 4 \\ &= 9.135 + j27.47 \\ &= 28.95 \angle 71.6^\circ \end{aligned}$$

$$I = \frac{12 \angle 0^\circ}{28.95 \angle 71.6^\circ} = 0.414 \angle -71.6^\circ$$

$$i(t) = 0.414 \cos(10t - 71.6^\circ) \text{ A}$$

Problem - 42

Calculate $i(t)$ in the circuit of Fig.

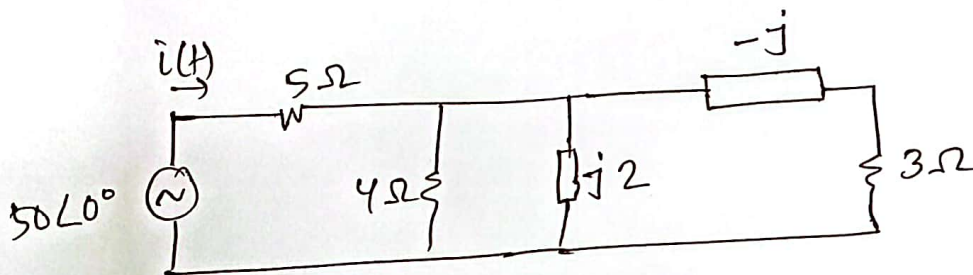


Solution: $\omega = 200 \text{ rad s}^{-1}$.

$$10 \text{ mH} \rightarrow Z_{10 \text{ mH}} = j\omega L = j \times 200 \times 10 \times 10^{-3} = j2 \Omega.$$

$$5 \text{ mF} \rightarrow Z_{5 \text{ mF}} = \frac{1}{j\omega C} = \frac{1}{j \times 200 \times 5 \times 10^{-3}} = -j \Omega.$$

$$50 \cos 200t \text{ V} = 50 \angle 0^\circ.$$



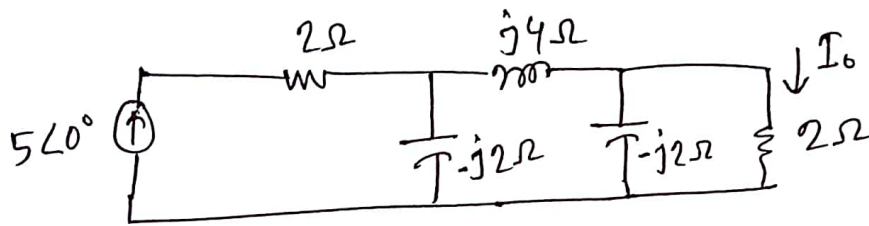
$$Z_T = ((3-j) \parallel j2 \parallel 4) + 5 = 6.189 + j0.865 = 11.22 \angle 4.42^\circ.$$

$$I = \frac{50 \angle 0^\circ}{11.22 \angle 4.42^\circ} = 4.46 \angle -4.42^\circ.$$

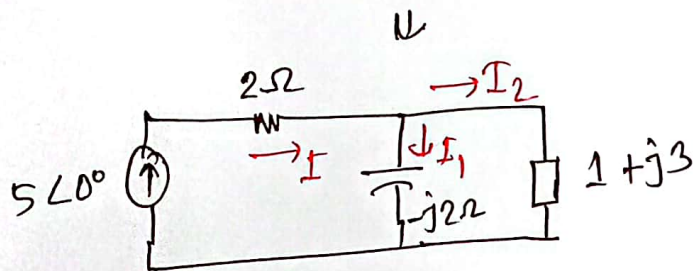
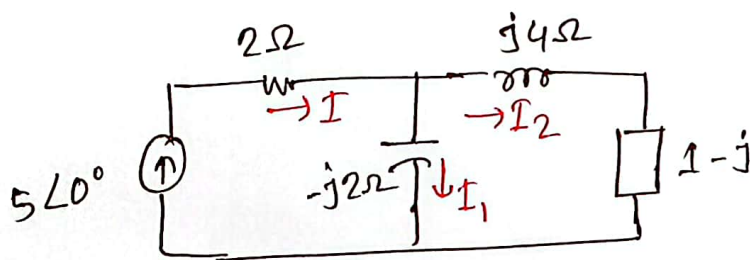
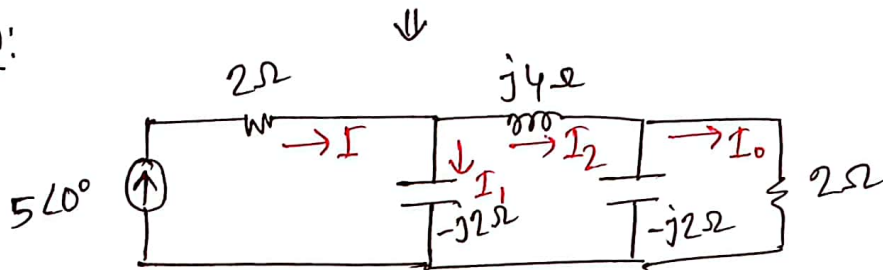
$$i(t) = 4.46 \cos(200t - 4.42^\circ)$$

Problem-43

Find current I_o in the network of Fig.



Solution:



$$Z_T = (-j2 \parallel (1+j3))$$

$$Z_T = (-j2 \parallel (1+j3)) + 2$$

$$= 4 - j4$$

$$= 5.656 \angle -45^\circ$$

Use current divider rule to find I_2

$$I_2 = \frac{j2 \parallel (1+j3)}{1+j3} \times I$$

$$= \frac{2-j4}{1+j3} \times 5 \angle 0^\circ$$

$$= 7.07 \angle -135^\circ.$$

Again, use current divider rule to find I_o .

$$I_o = \frac{-j2 \parallel 2}{2} \times I_2$$

$$= \frac{1-j}{2} \times 7.07 \angle -135^\circ$$

$$= -4.99 \text{ A}$$

$$I_o = -4.99 \text{ A}.$$